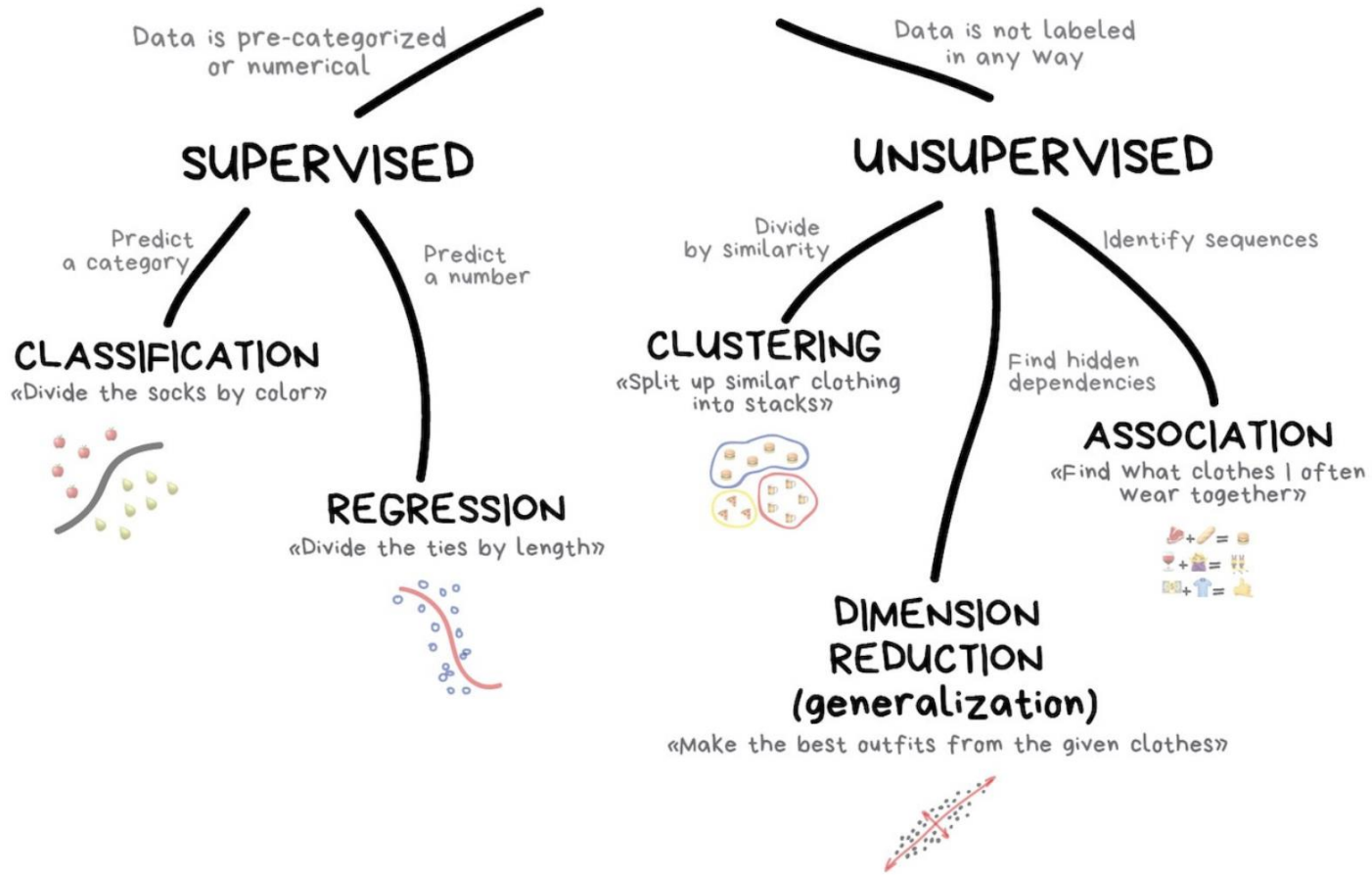
Data Clustering

Assoc. Prof. Kitsana Waiyamai, Ph. D.
Dept. Of Computer Engineering
Faculty of Engineering, Kasetsart University
Bangkok, Thailand



Classical Machine Learning



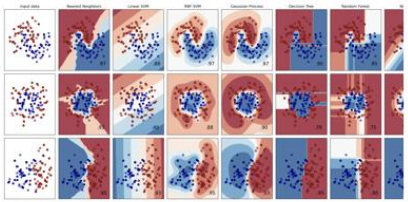
https://vas3k.com/blog/machine_learning/?ref=hn

Scikit-Learn: https://scikit-learn.org

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.
Algorithms: Gradient boosting, nearest neighbors, random forest, logistic regression, and more...

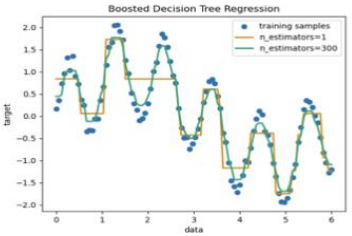


Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, Stock prices.
Algorithms: Gradient boosting, nearest neighbors, random forest, ridge, and more...

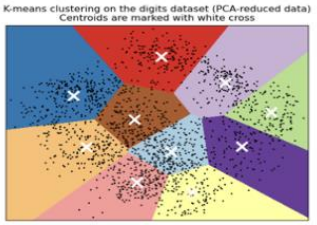


Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, Grouping experiment outcomes
Algorithms: k-Means, HDBSCAN, hierarchical clustering, and more...

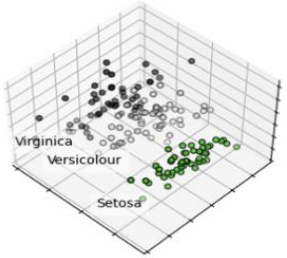


Examples

Dimensionality reduction

Reducing the number of random variables to consider.

Applications: Visualization, Increased efficiency
Algorithms: PCA, feature selection, non-negative matrix factorization, and more...

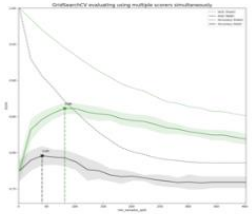


Examples

Model selection

Comparing, validating and choosing parameters and models.

Applications: Improved accuracy via parameter tuning
Algorithms: grid search, cross validation, metrics, and more...

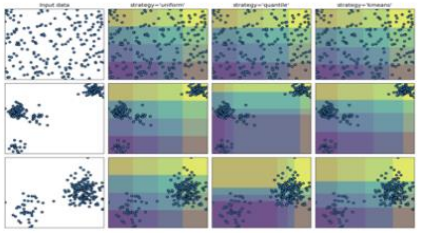


Examples

Preprocessing

Feature extraction and normalization.

Applications: Transforming input data such as text for use with machine learning algorithms.
Algorithms: preprocessing, feature extraction, and more...



Examples

Data Clustering (การแบ่งกลุ่มข้อมูล)

- Provides a way to learn about the structure of complex data.
- Simply finds structure that exists in the data without regard to any target variable => **Un-supervised learning**.

- Data classification organizes data in given classes based on attribute values => **Supervised-learning** !
- **Data Clustering** divides data into meaningful or useful clusters, without given classes => **Un-Supervised-learning** !

- **Data Clustering** is an un-supervised learning technique to form **groups (clusters) of similar records**.

Data Clustering Applications

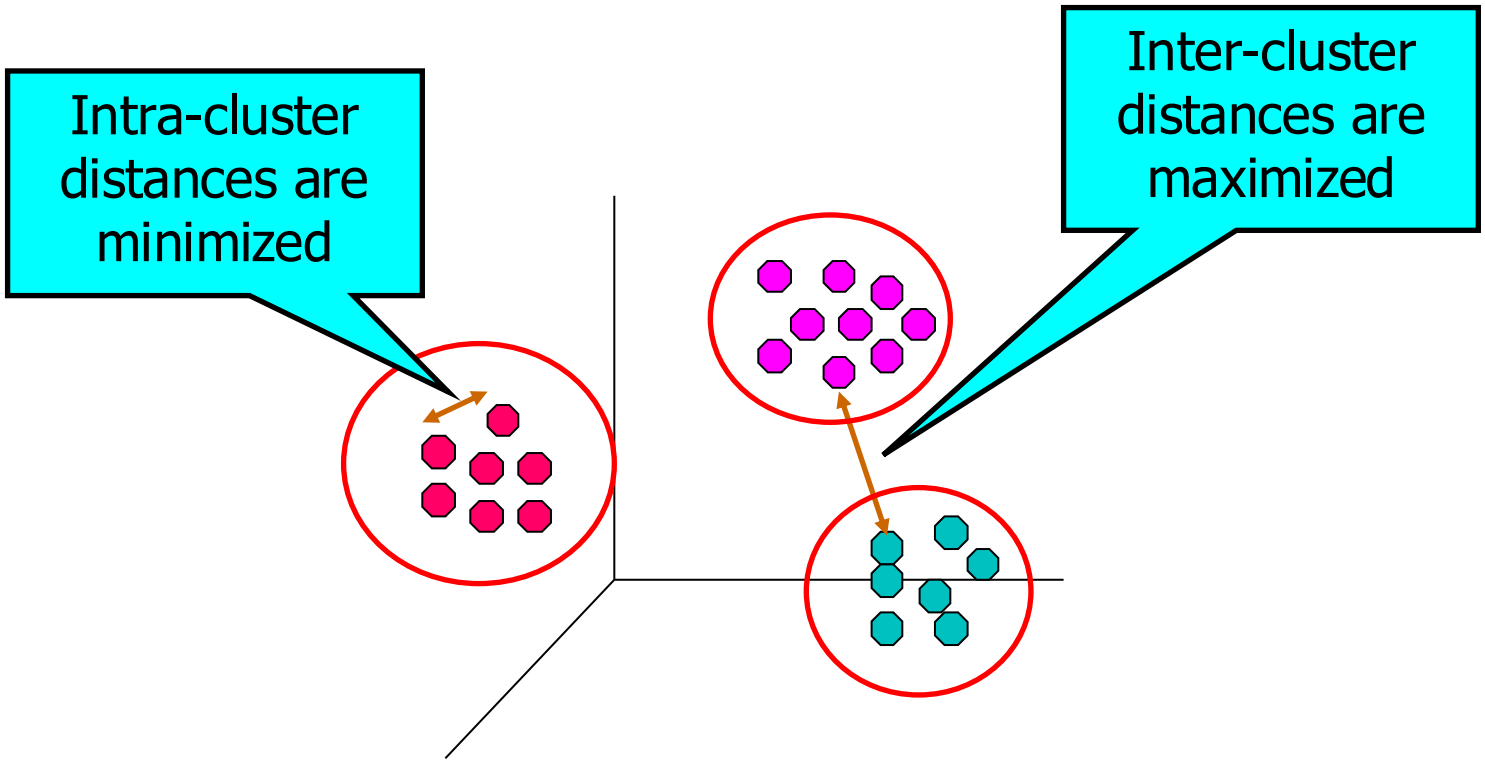
- **Applications**:
  - Customer segmentation
  - Recommendation systems
  - Grouping electoral voters in election campaigning
  - Clustering of user queries
  - Etc.

Un-supervised Learning Techniques

- Association Rules
  - What happens together?
- **Data Clustering**
  - **What belongs together?**

What is Data Clustering?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Data Clustering

- Cluster: A collection of data objects
  - similar (or related) to one another within the same group
  - dissimilar (or unrelated) to the objects in other groups
- Clustering = set of clusters
- Unsupervised learning: no predefined classes
- Typical applications
  - As a pre-processing step for other algorithms
  - As a stand-alone tool to get insight into data distribution

Example: Public Utilities

Goal: find clusters of similar utilities

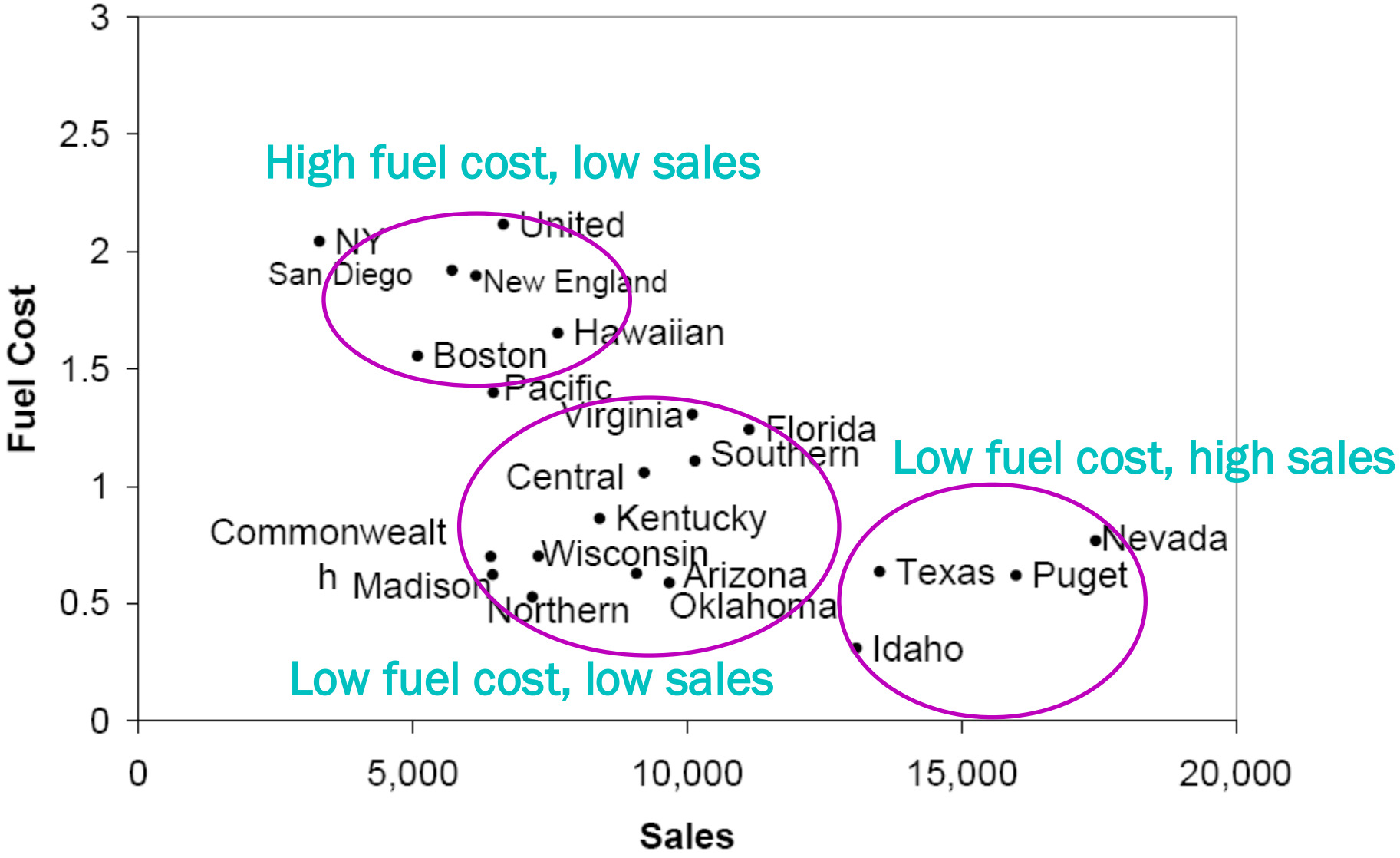
Data: 22 firms, 8 variables

- Fixed-charge covering ratio
- Rate of return on capital
- Cost per kilowatt capacity
- Annual load factor
- Growth in peak demand
- Sales
- % nuclear
- Fuel costs per kwh

Example: 22 Public Utilities

Company	Fixed_charge	RoR	Cost	Load	Δ Demand	Sales	Nuclear	Fuel_Cost
Arizona	1.06	9.2	151	54.4	1.6	9077	0	0.628
Boston	0.89	10.3	202	57.9	2.2	5088	25.3	1.555
Central	1.43	15.4	113	53	3.4	9212	0	1.058
Commonwealth	1.02	11.2	168	56	0.3	6423	34.3	0.7
Con Ed NY	1.49	8.8	192	51.2	1	3300	15.6	2.044
Florida	1.32	13.5	111	60	-2.2	11127	22.5	1.241
Hawaiian	1.22	12.2	175	67.6	2.2	7642	0	1.652
Idaho	1.1	9.2	245	57	3.3	13082	0	0.309
Kentucky	1.34	13	168	60.4	7.2	8406	0	0.862
Madison	1.12	12.4	197	53	2.7	6455	39.2	0.623
Nevada	0.75	7.5	173	51.5	6.5	17441	0	0.768
New England	1.13	10.9	178	62	3.7	6154	0	1.897
Northern	1.15	12.7	199	53.7	6.4	7179	50.2	0.527
Oklahoma	1.09	12	96	49.8	1.4	9673	0	0.588
Pacific	0.96	7.6	164	62.2	-0.1	6468	0.9	1.4
Puget	1.16	9.9	252	56	9.2	15991	0	0.62
San Diego	0.76	6.4	136	61.9	9	5714	8.3	1.92
Southern	1.05	12.6	150	56.7	2.7	10140	0	1.108
Texas	1.16	11.7	104	54	-2.1	13507	0	0.636
Wisconsin	1.2	11.8	148	59.9	3.5	7287	41.1	0.702
United	1.04	8.6	204	61	3.5	6650	0	2.116
Virginia	1.07	9.3	174	54.3	5.9	10093	26.6	1.306

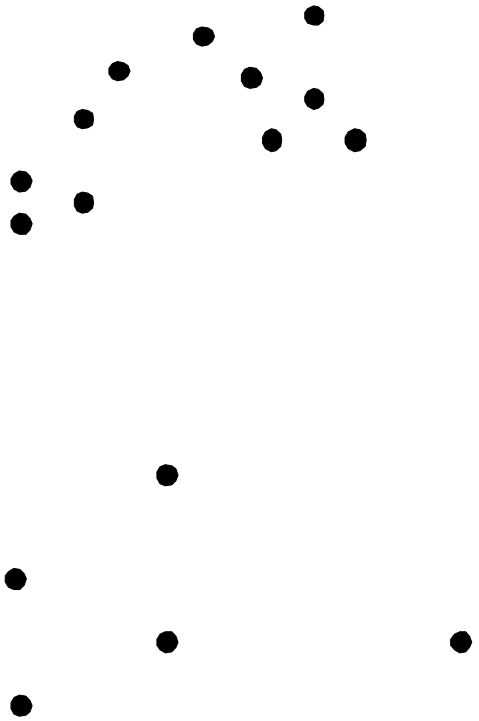
Sales & Fuel Cost: 3 rough clusters can be seen



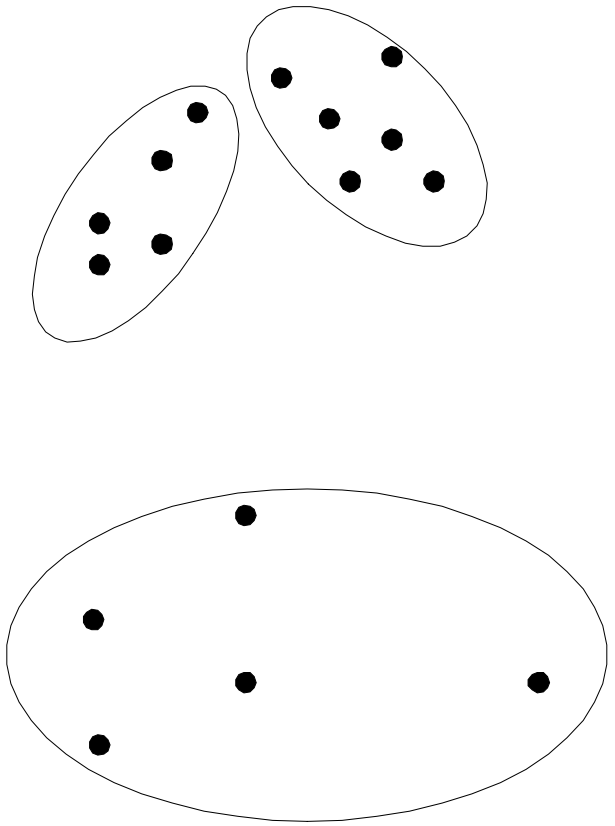
Types of Clustering Techniques

- A clustering is a set of clusters
- Important distinction between hierarchical and partitional sets of clusters
- Partitional clustering
  - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
- Hierarchical clustering
  - A set of nested clusters organized as a hierarchical tree
- Density-based Clustering
- Etc.

Partitional Clustering

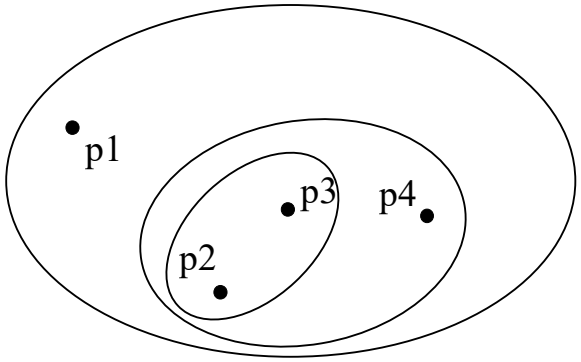


Original Points

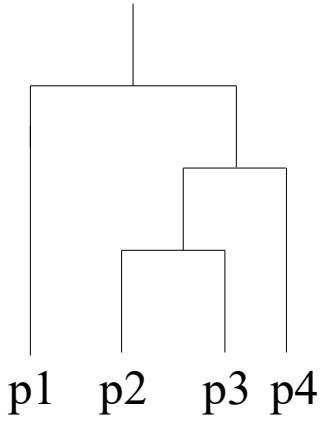


A Partitional Clustering

Hierarchical Clustering



Hierarchical Clustering



Dendrogram

Notion of a Cluster can be Ambiguous



How many clusters?



Six Clusters



Two Clusters



Four Clusters

Partitioning Algorithms: Basic Concept

Given a *k*, find a partition of *k clusters* that optimizes the chosen partitioning criterion

- Global optimal: exhaustively enumerate all partitions
- Heuristic methods: *k-means* and *k-medoids* algorithms
- *k-means* (MacQueen'67): Each cluster is represented by the center of the cluster
- *k-medoids* or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

- K-Means: Construct a partition of a database ***D*** of ***n*** objects into a set of ***k*** clusters, s.t., min sum of squared distance

$$E = \Sigma_{i=1}^{k} \Sigma_{p \in C_i} (p - m_i)^2$$

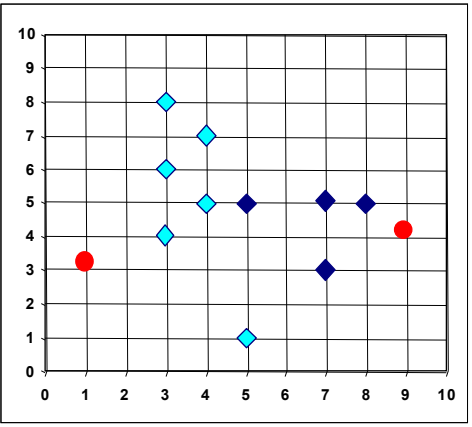
Simple Clustering: K-means

- Uses a set of clusters(K), then iteratively recalculates the center of each cluster.
- Works with numeric data only.

1) Pick a number (K) of cluster centers (at random)
2) Assign every item to its nearest cluster center (example, using Euclidean distance)
3) Move each cluster center to the mean of its assigned items
4) Repeat steps 2,3 until convergence (or change in cluster assignments less than a threshold)

The *K-Means* Clustering Method

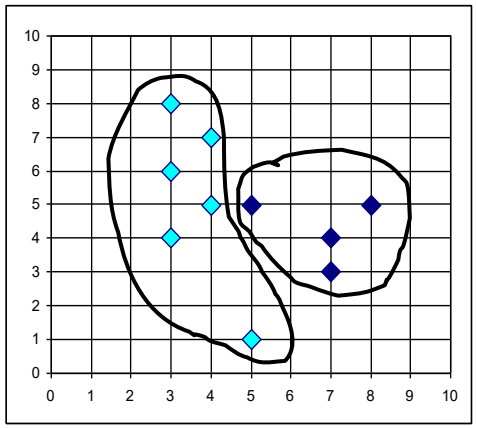
- Example



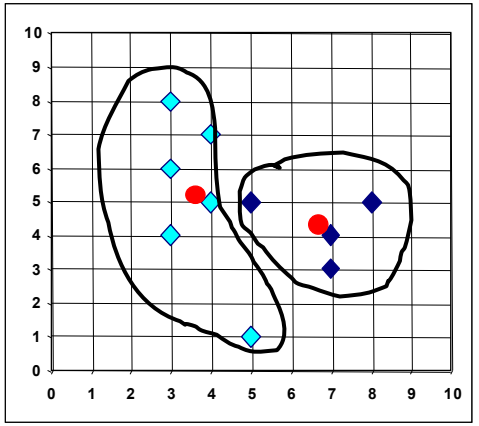
K=2

Arbitrarily choose K object as initial cluster center

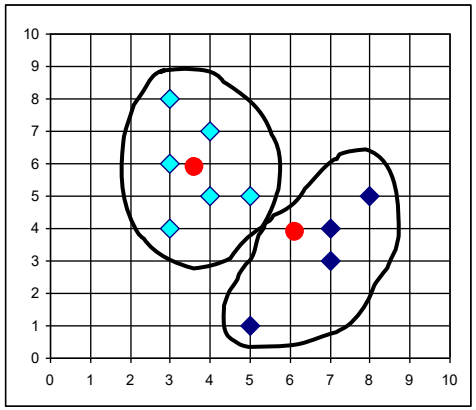
Assign each objects to most similar center



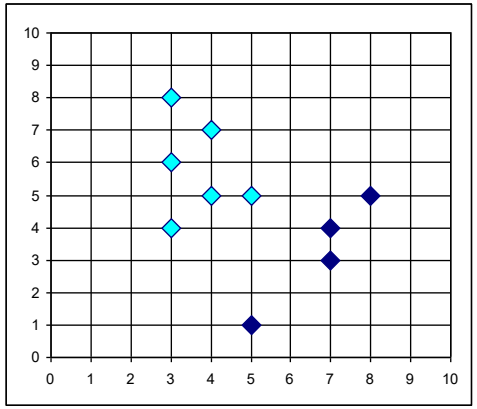
Update the cluster means



reassign

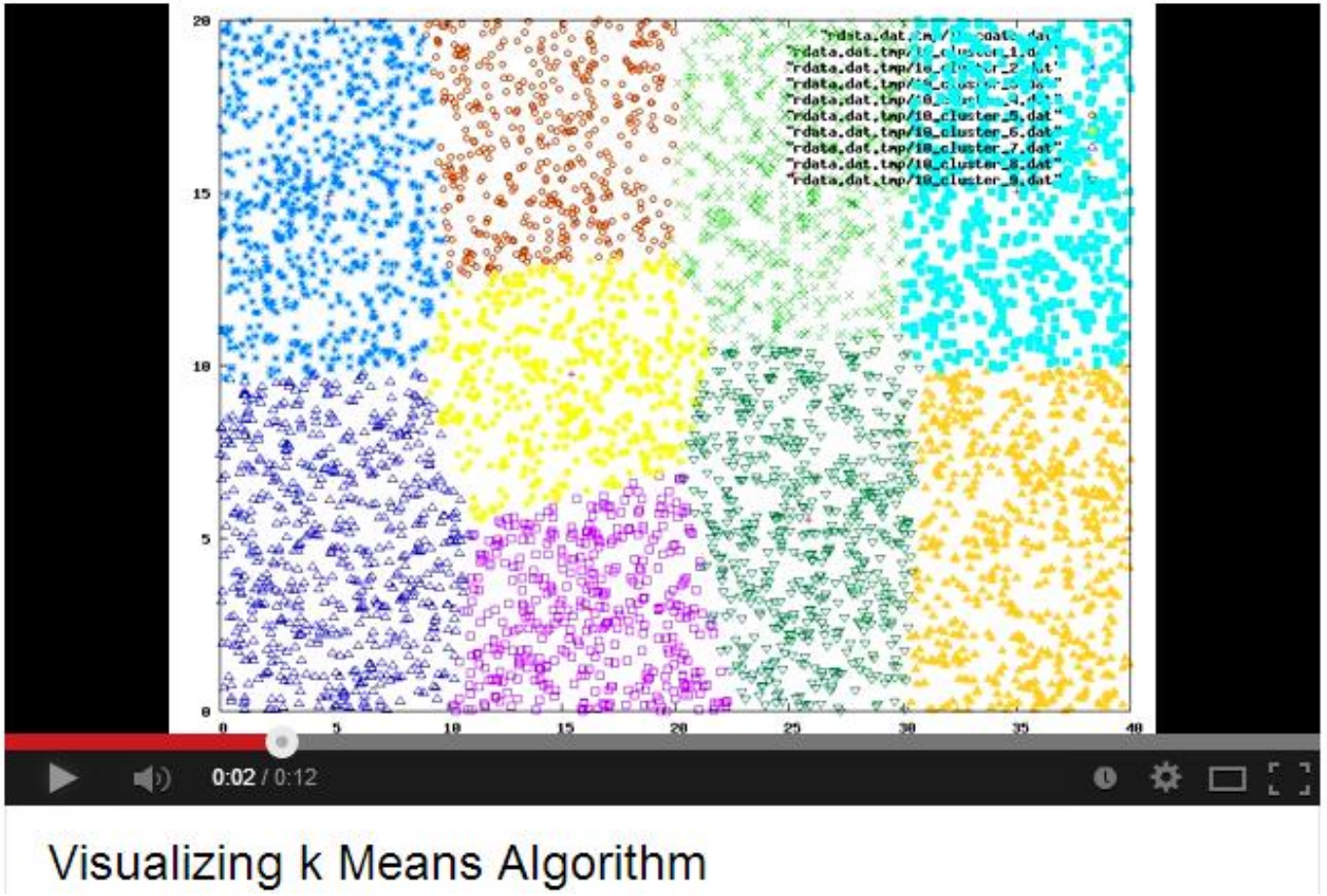


Update the cluster means



reassign

K-mean clustering visualization



http://www.youtube.com/watch?v=gSt4_kcZPxE

Simple Clustering: K-means

Not changed

- Data points

Changed

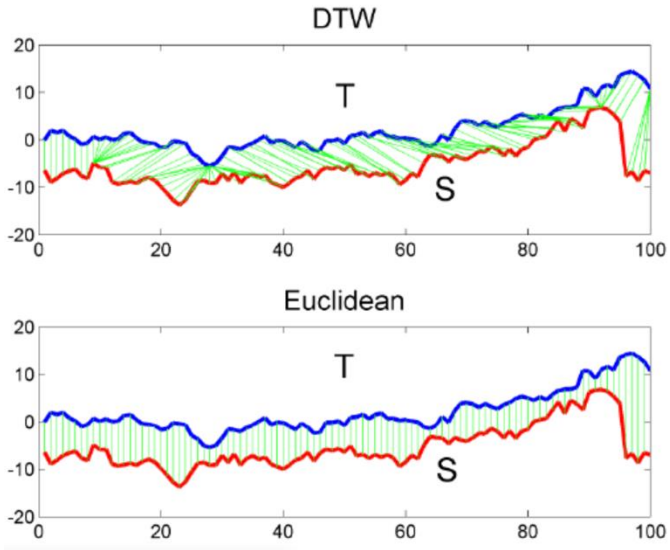
- Boundary
- Centroid
- Membership

Distance Between Two Records

Euclidean Distance is most popular:

$$d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2}$$

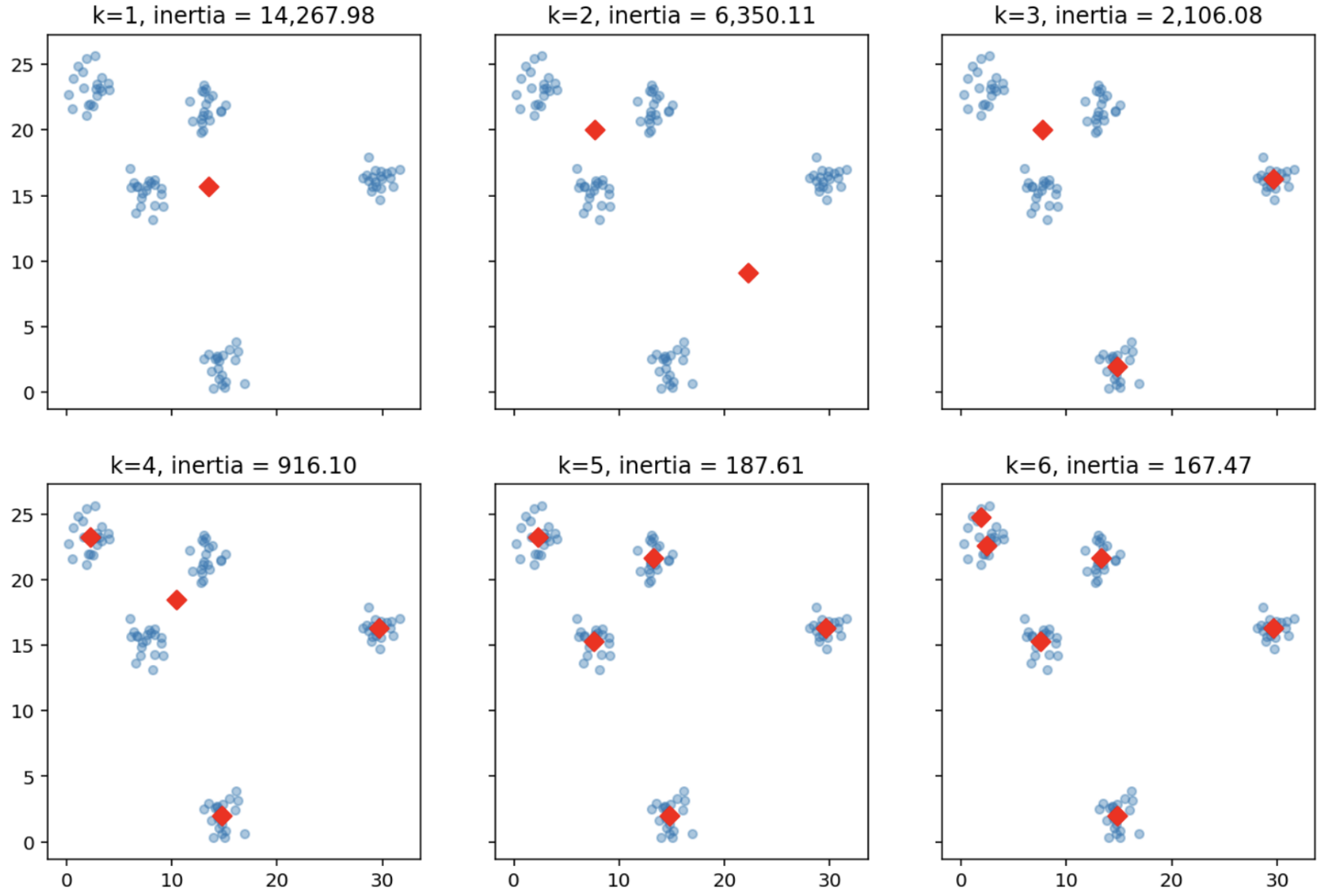
- **Problem:** Raw distance measures are highly influenced by scale of measurements
- **Solution:** normalize (standardize) the data first
  - Subtract mean, divide by std. deviation
  - Also called **z-scores**
- **Other Distances**
  - Correlation-based similarity
  - Jacccard coeffient categorical
  - [Continuous + categorical] Gower's similarity
  - Text-Analytics => Angle [lot of words]
  - Time series => DTW
  - Etc.



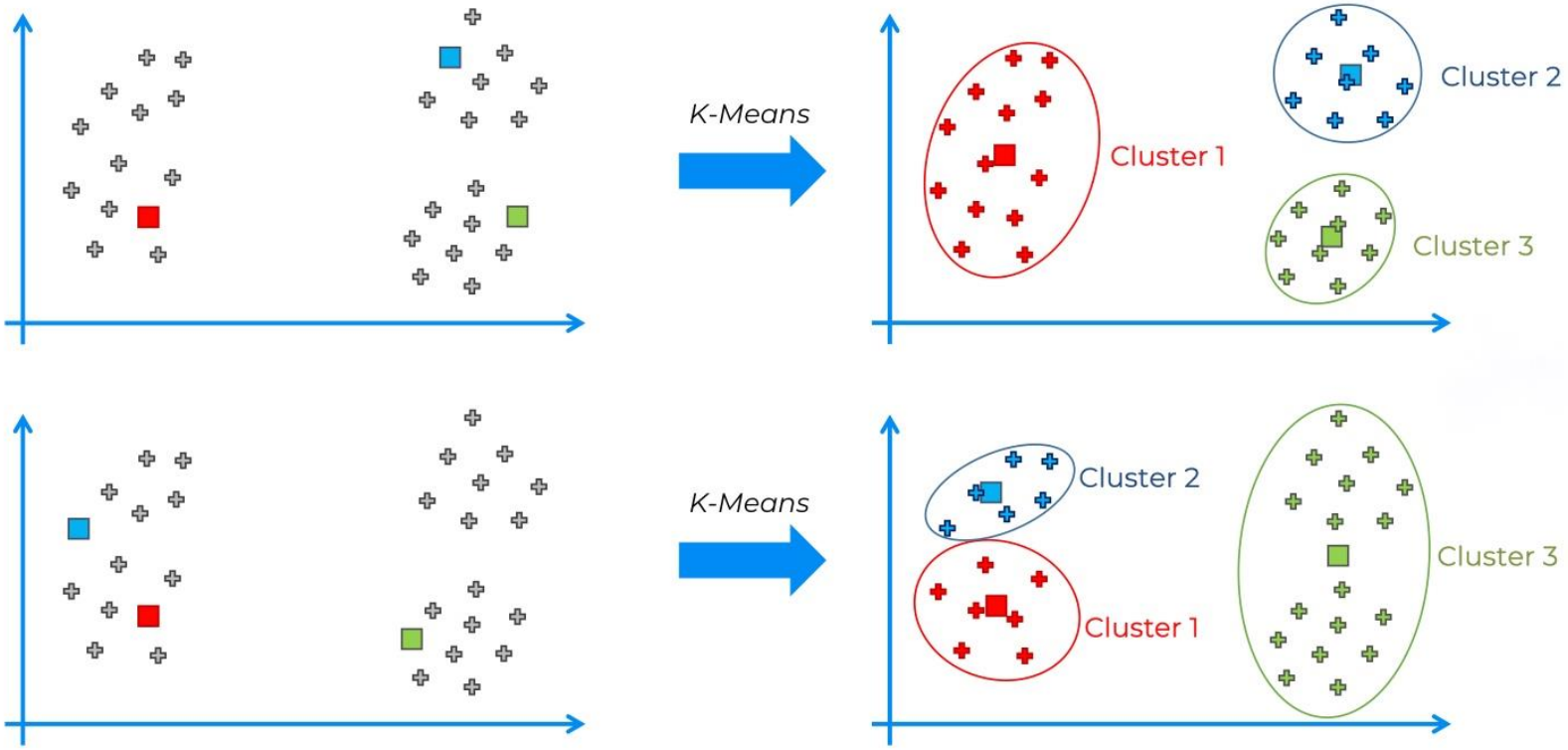
Pre-processing and Post-processing

- Pre-processing
  - Normalize the data
  - Dimension reduction
  - Eliminate outliers
- Post-processing
  - Eliminate small clusters that may represent outliers
  - Split ‘loose’ clusters,
  - Merge clusters that are ‘close’
  - Etc.

K-Mean: K-value ?



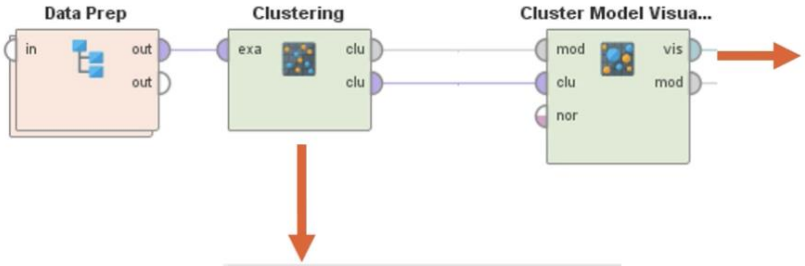
Some Problem(s) of K-Means



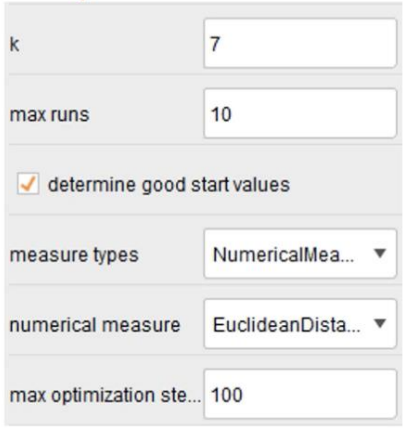
Result can vary significantly depending on initial choice of seeds

Use **K-Mean++** [David Arthur and Sergei Vassilvitskii 2007] to determine good start seeds.

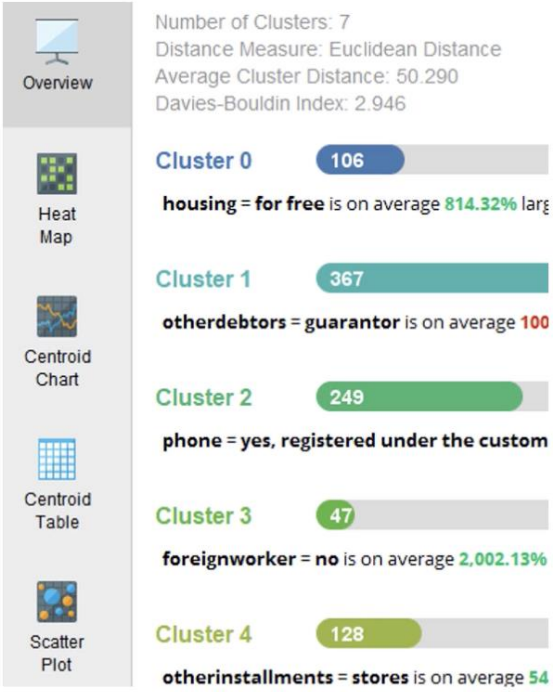
RapidMiner K-Mean Clustering



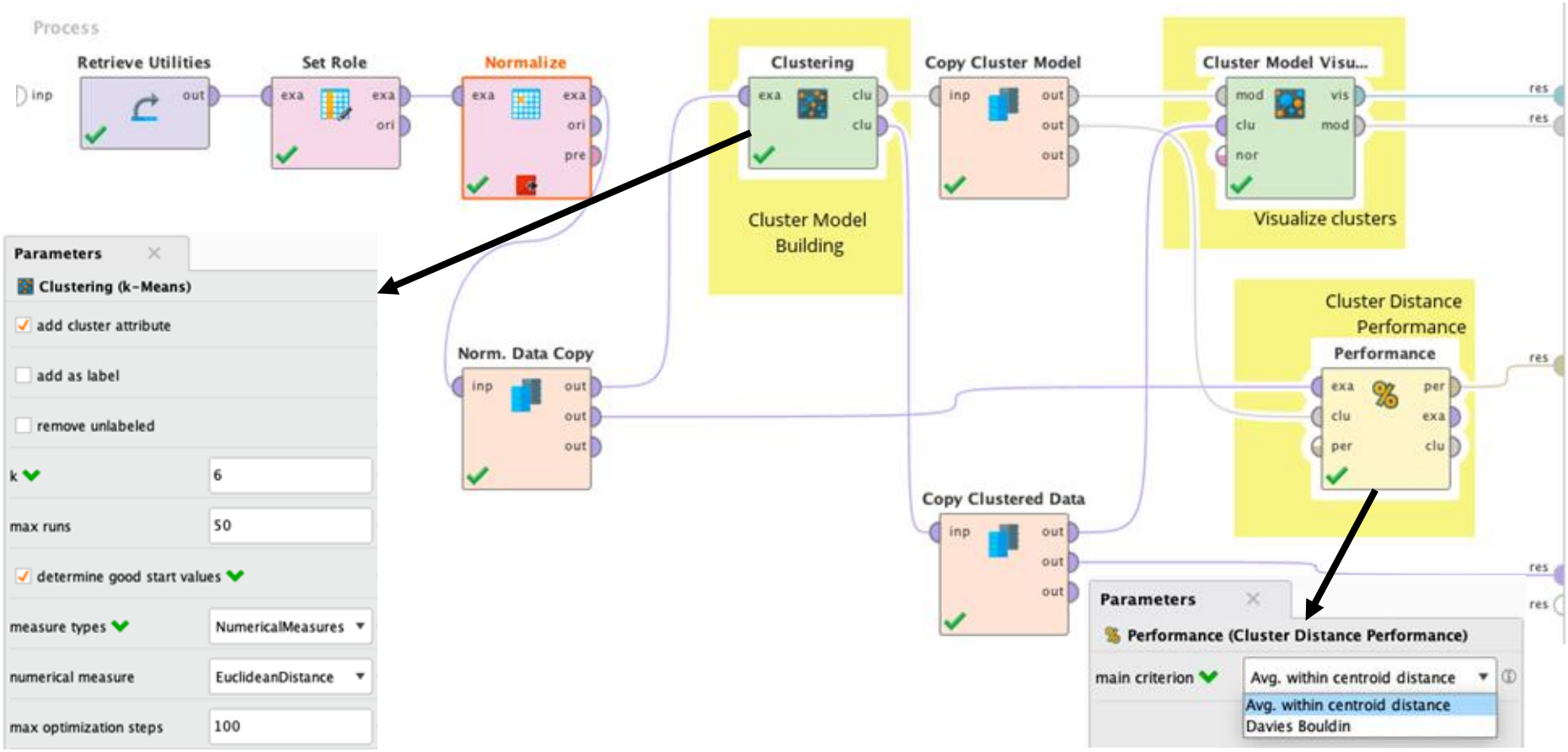
k sets number of clusters



Measure types sets how distances are measured



K-means clustering in RapidMiner



Rapidminer K-means results

ExampleSet (Copy Clustered Data)

Open in Turbo Prep | Auto Model

Filter (22 / 22 examples): all

Row No.	Company	cluster	Fixed_charge	RoR	Cost	Load_factor	Demand_gr...	Sales	Nuclear	Fuel_Cost
7	Hawaiian	cluster_0	0.574	0.652	0.166	2.381	-0.334	-0.358	-0.715	0.988
9	Kentucky	cluster_0	1.224	1.009	-0.004	0.767	1.270	-0.143	-0.715	-0.433
12	New England	cluster_0	0.086	0.073	0.238	1.126	0.147	-0.777	-0.715	1.428
15	Pacific	cluster_0	-0.835	-1.398	-0.102	1.171	-1.071	-0.689	-0.661	0.535
21	United	cluster_0	-0.402	-0.952	0.870	0.902	0.083	-0.638	-0.715	1.822
2	Boston	cluster_1	-1.215	-0.194	0.821	0.207	-0.334	-1.078	0.792	0.813
4	Commonwe...	cluster_1	-0.510	0.207	-0.004	-0.219	-0.943	-0.702	1.328	-0.724
10	Madison	cluster_1	0.032	0.741	0.700	-0.892	-0.173	-0.693	1.620	-0.863
13	Northern	cluster_1	0.195	0.875	0.748	-0.735	1.013	-0.489	2.275	-1.035
20	Wisconsin	cluster_1	0.466	0.474	-0.490	0.655	0.083	-0.458	1.733	-0.721
22	Virginia	cluster_1	-0.239	-0.640	0.141	-0.600	0.853	0.332	0.869	0.366
1	Arizona	cluster_2	-0.293	-0.685	-0.417	-0.578	-0.526	0.046	-0.715	-0.854
3	Central	cluster_2	1.712	2.078	-1.340	-0.892	0.051	0.084	-0.715	-0.080
6	Florida	cluster_2	1.116	1.232	-1.388	0.678	-1.745	0.623	0.625	0.249
14	Oklahoma	cluster_2	-0.131	0.563	-1.752	-1.609	-0.590	0.214	-0.715	-0.926
18	Southern	cluster_2	-0.347	0.830	-0.441	-0.062	-0.173	0.345	-0.715	0.009
19	Texas	cluster_2	0.249	0.429	-1.558	-0.667	-1.713	1.294	-0.715	-0.839
5	NY	cluster_3	2.037	-0.863	0.578	-1.295	-0.719	-1.581	0.214	1.693
8	Idaho	cluster_4	-0.076	-0.685	1.865	0.005	0.019	1.174	-0.715	-1.427
11	Nevada	cluster_4	-1.973	-1.442	0.117	-1.228	1.045	2.402	-0.715	-0.602
16	Puget	cluster_4	0.249	-0.373	2.035	-0.219	1.911	1.994	-0.715	-0.868
17	San Diego	cluster_5	-1.919	-1.932	-0.781	1.103	1.847	-0.901	-0.220	1.470

Result History | Cluster Model (Clustering)

Description

Folder View

Cluster Model

```
Cluster 0: 5 items
Cluster 1: 6 items
Cluster 2: 6 items
Cluster 3: 1 items
Cluster 4: 3 items
Cluster 5: 1 items
Total number of items: 22
```

Result History | Cluster Model (Clustering)

Description

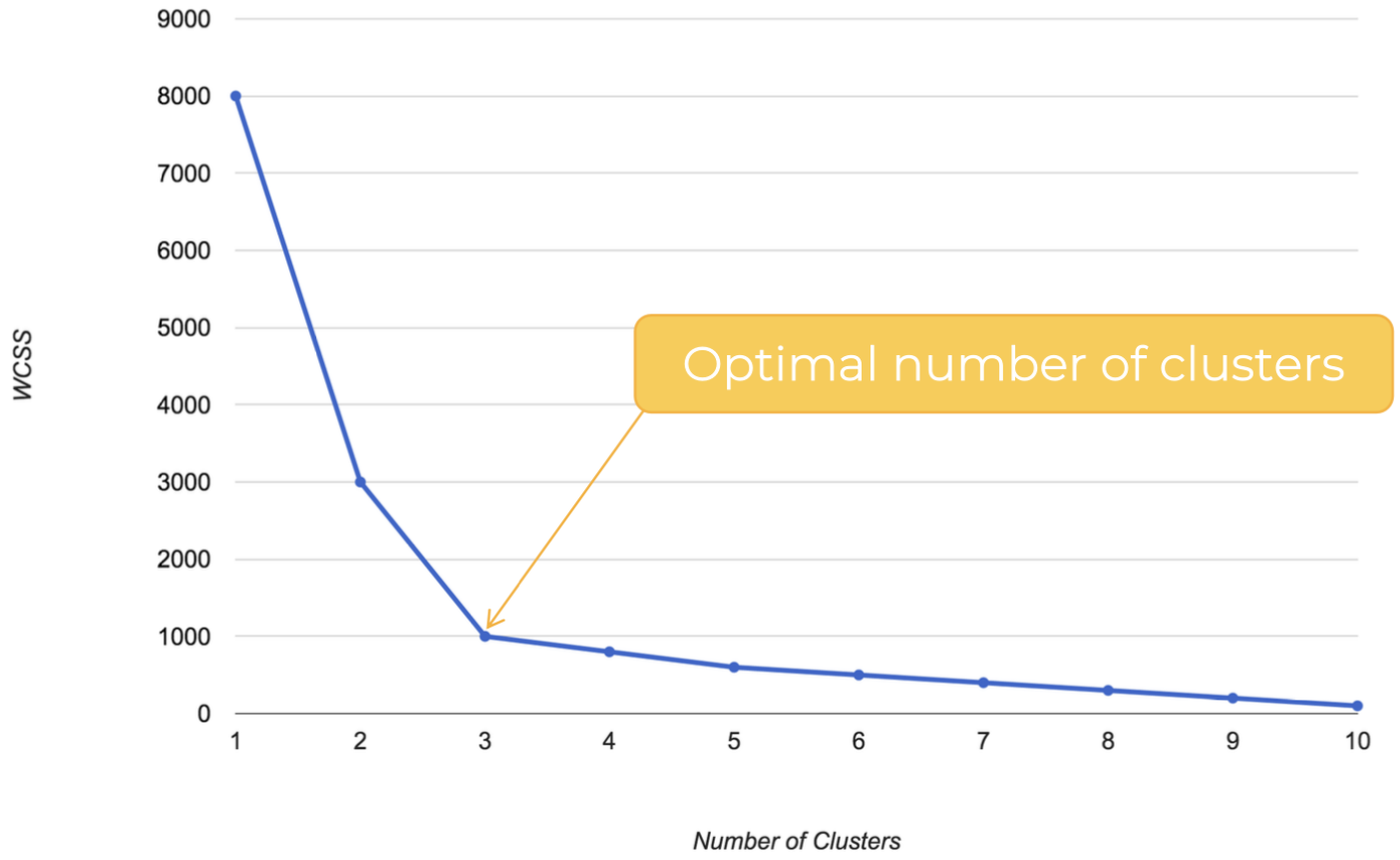
Folder View

Graph

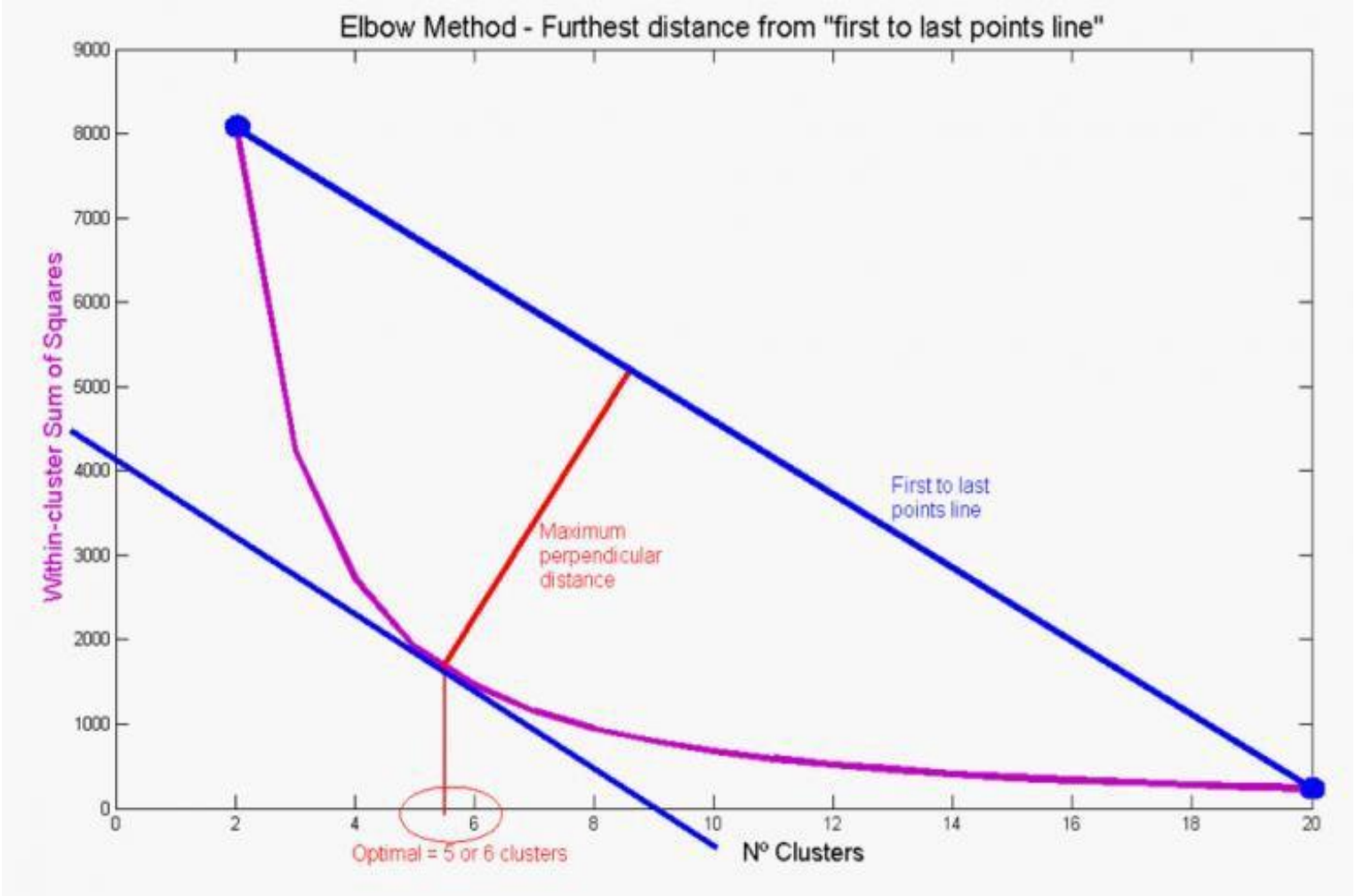
- root
  - cluster_0
    - Hawaiian
    - Kentucky
    - New England
    - Pacific
    - United
  - cluster_1
    - Boston
    - Commonwealth
    - Madison
    - Northern
    - Wisconsin
    - Virginia
  - cluster_2
    - Arizona
    - Central
    - Florida
    - Oklahoma
    - Southern
    - Texas
  - cluster_3
    - NY
  - cluster_4
    - Idaho
    - Nevada
    - Puget
  - cluster_5
    - San Diego

How to know K: Elbow method

- First, draw relationship between the number of clusters and Within Cluster Sum of Squares (WCSS),
- Then we select the number of clusters where the change in WCSS begins to level off.



How to know K: Elbow method



Python: K-Means

```
# K-Means Clustering
# Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

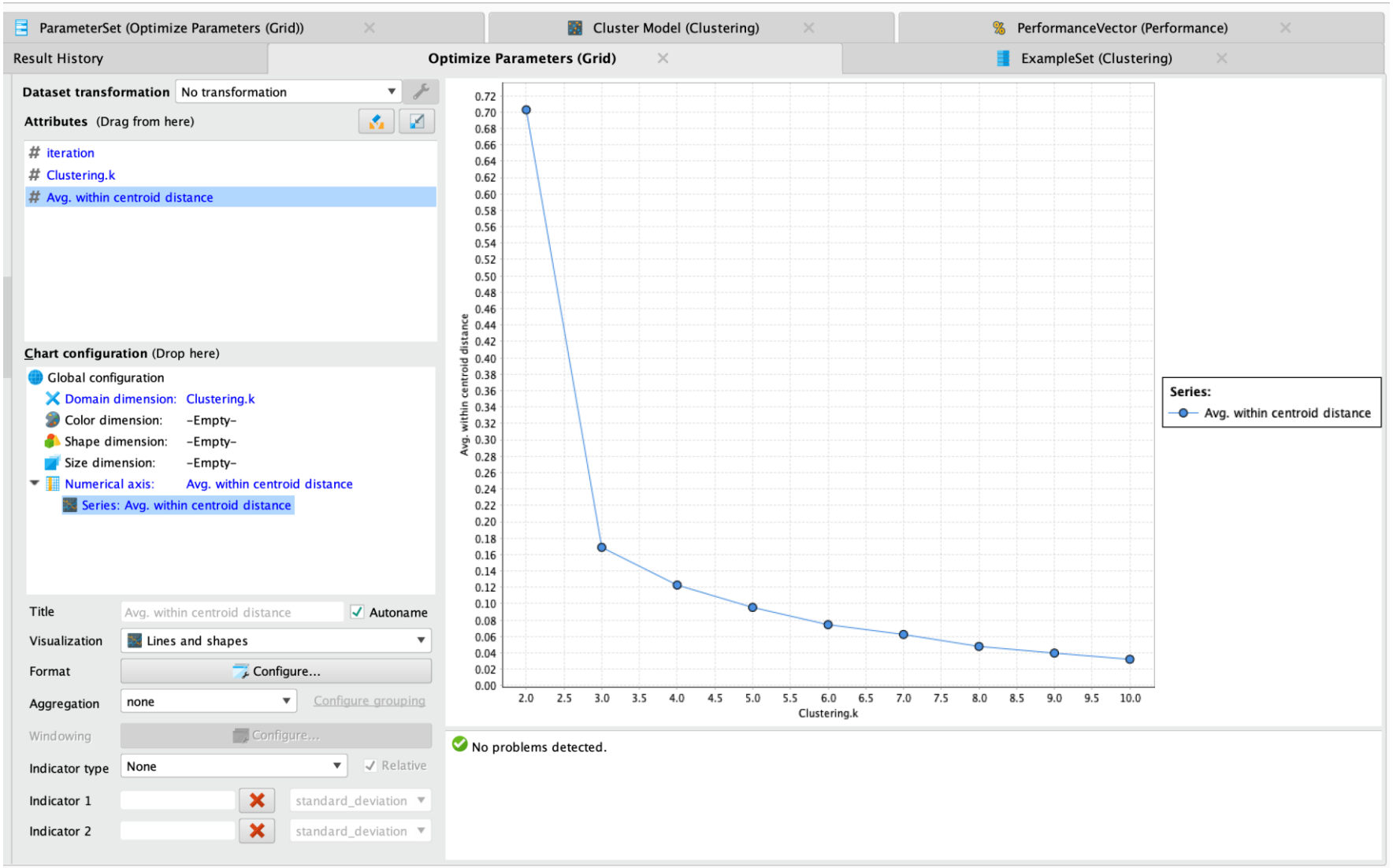
# Importing the dataset
dataset = pd.read_csv('Mall_Customers.csv')
X = dataset.iloc[:, [3, 4]].values
# y = dataset.iloc[:, 3].values

# Using the elbow method to find the optimal number of clusters
from sklearn.cluster import KMeans
wcss = []
for i in range(2, 11):
    kmeans = KMeans(n_clusters = i, init = 'k-means++', random_state = 0)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)
plt.plot(range(2, 11), wcss)
plt.title('The Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()

# Fitting K-Means to the dataset
kmeans = KMeans(n_clusters = 5, init = 'k-means++', random_state = 0)
y_kmeans = kmeans.fit_predict(X)

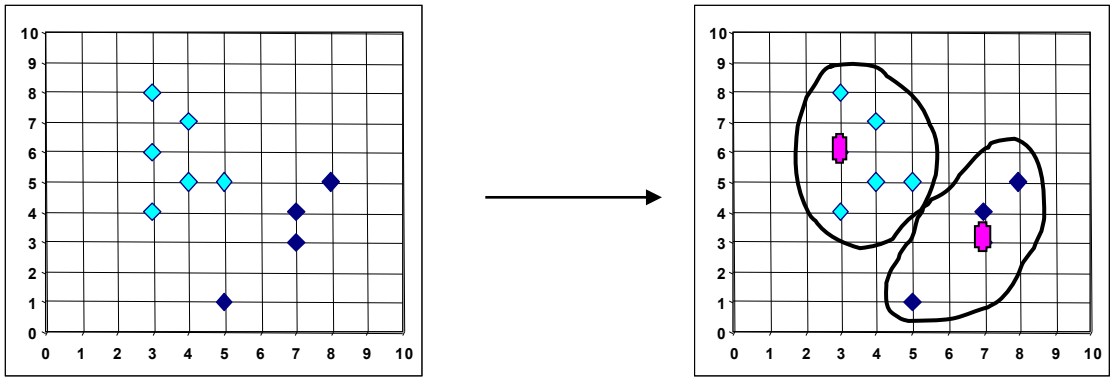
# Visualising the clusters
plt.scatter(X[y_kmeans == 0, 0], X[y_kmeans == 0, 1], s = 100, c = 'red', label = 'Cluster 1')
plt.scatter(X[y_kmeans == 1, 0], X[y_kmeans == 1, 1], s = 100, c = 'blue', label = 'Cluster 2')
plt.scatter(X[y_kmeans == 2, 0], X[y_kmeans == 2, 1], s = 100, c = 'green', label = 'Cluster 3')
plt.scatter(X[y_kmeans == 3, 0], X[y_kmeans == 3, 1], s = 100, c = 'cyan', label = 'Cluster 4')
plt.scatter(X[y_kmeans == 4, 0], X[y_kmeans == 4, 1], s = 100, c = 'magenta', label = 'Cluster 5')
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s = 300, c = 'yellow', label = 'Centroids')
plt.title('Clusters of customers')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```

How to know K: Elbow method

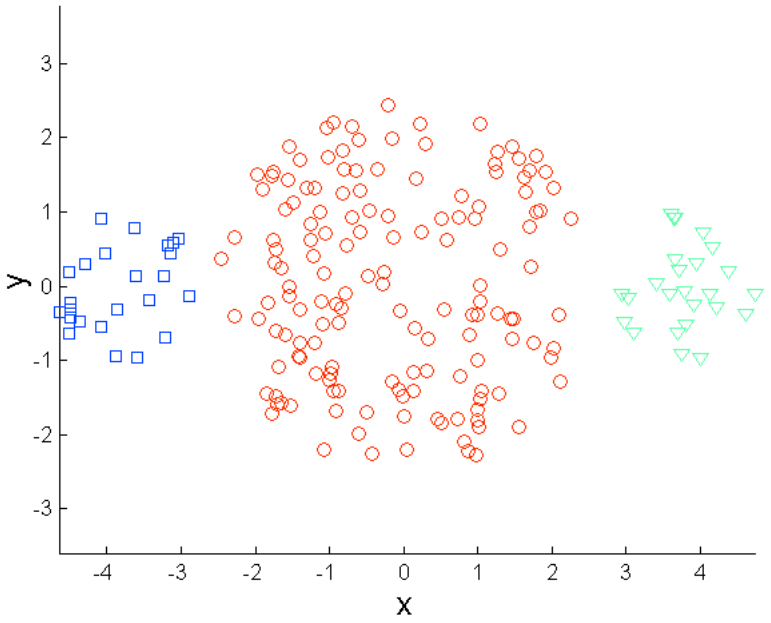


K-means variations

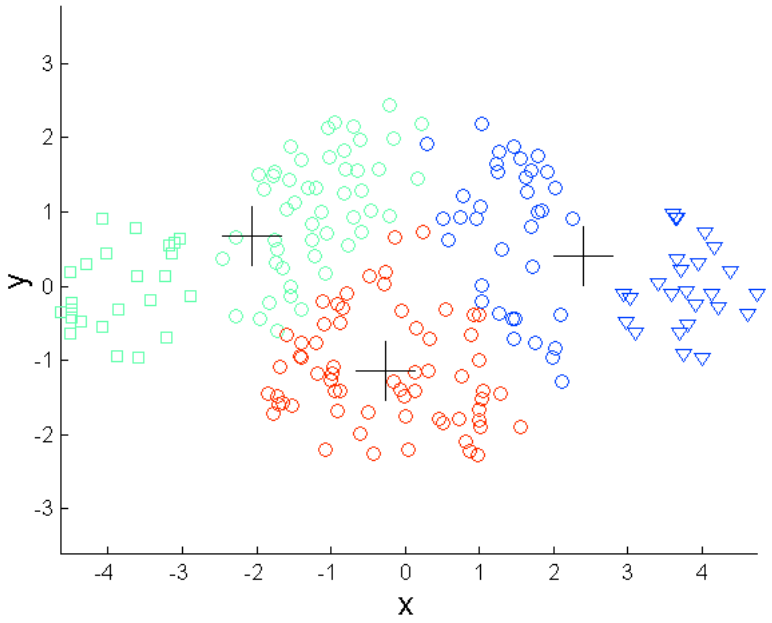
- The k-means algorithm is sensitive to outliers !
  - Since an object with an extremely large value may substantially distort the distribution of the data.
  - Mean of 1, 3, 5, 7, 1009 is
  - Median of 1, 3, 5, 7, 1009 is
  - Median advantage: not affected by extreme values
- K-Medoids or PAM: Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster.
- Etc.



Limitations of K-means: Differing Sizes

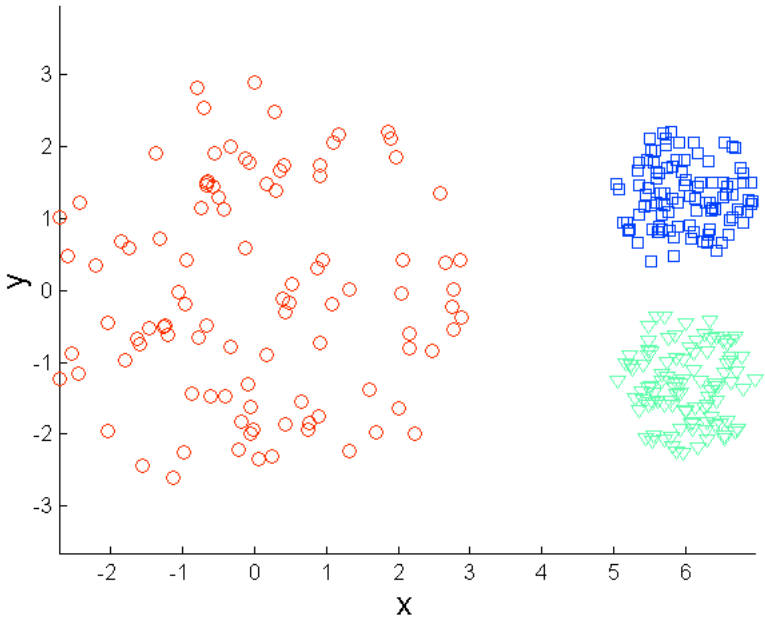


Original Points

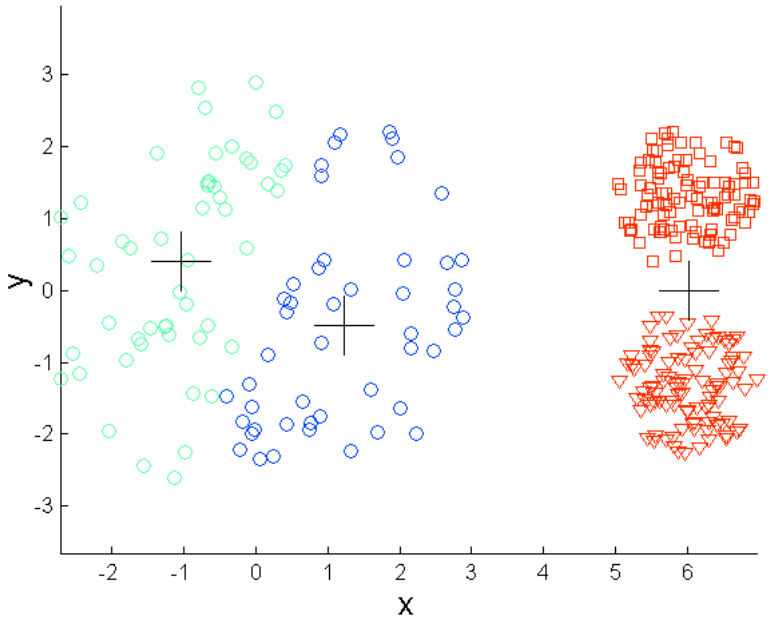


K-means (3 Clusters)

Limitations of K-means: Differing Density

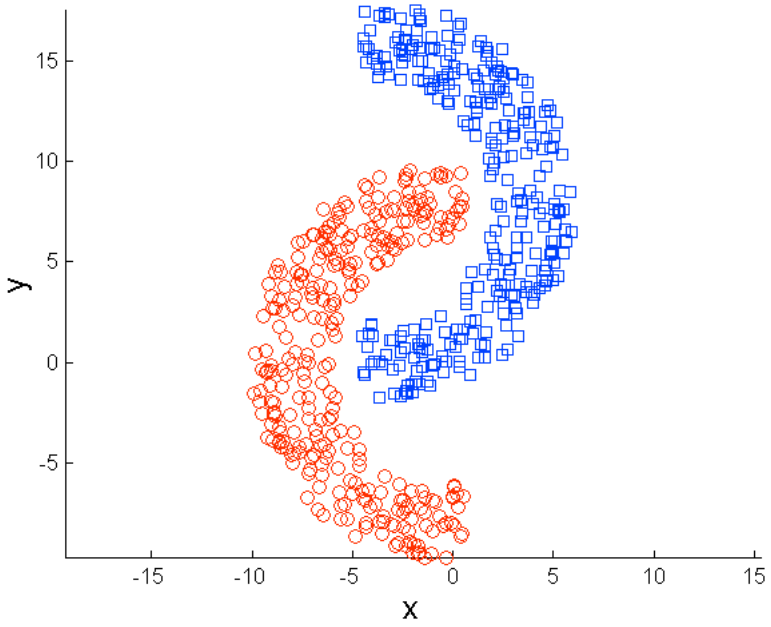


Original Points

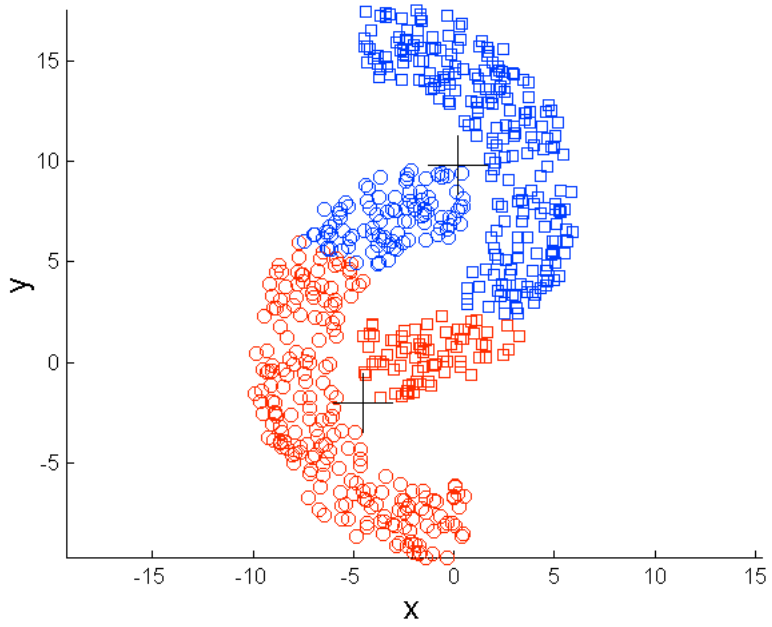


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes

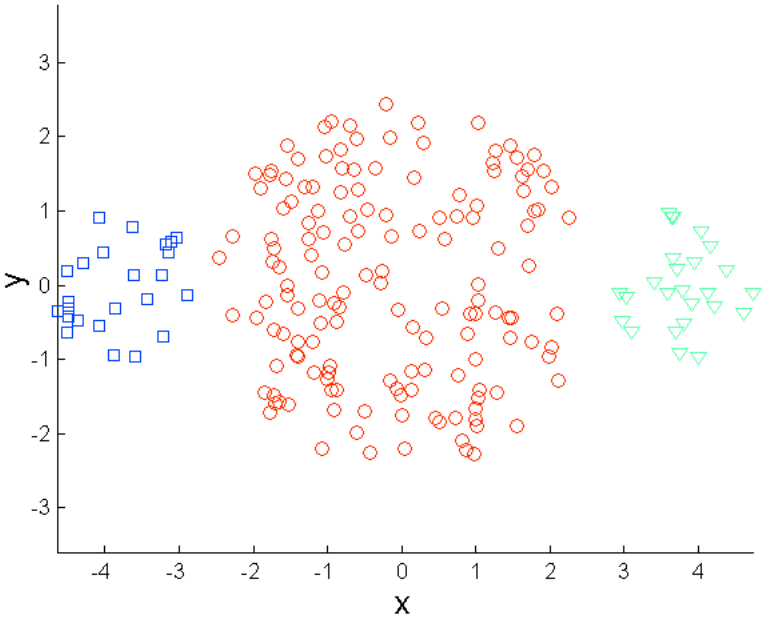


Original Points

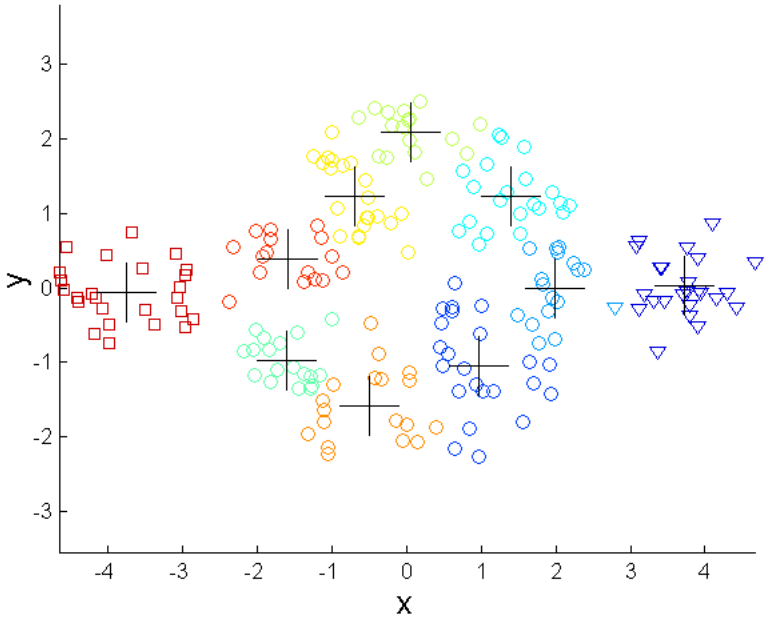


K-means (2 Clusters)

Overcoming K-means Limitations



Original Points

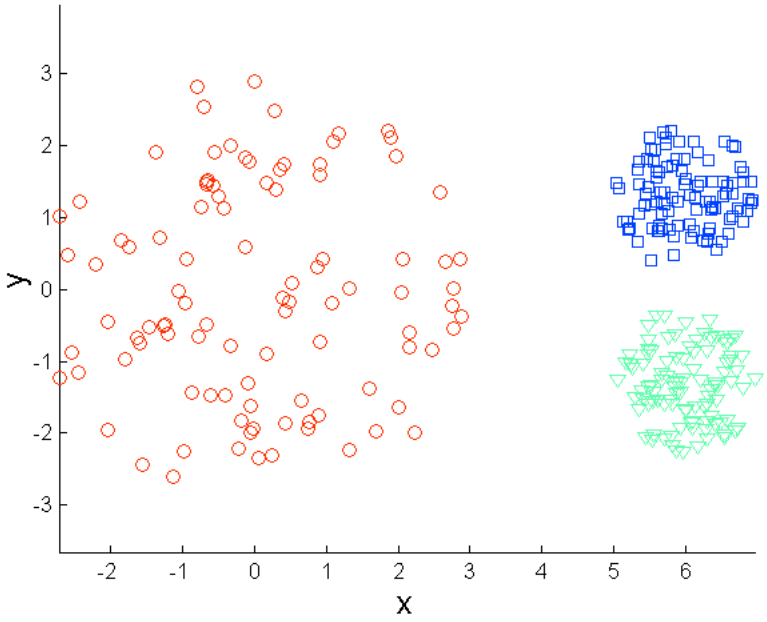


K-means Clusters

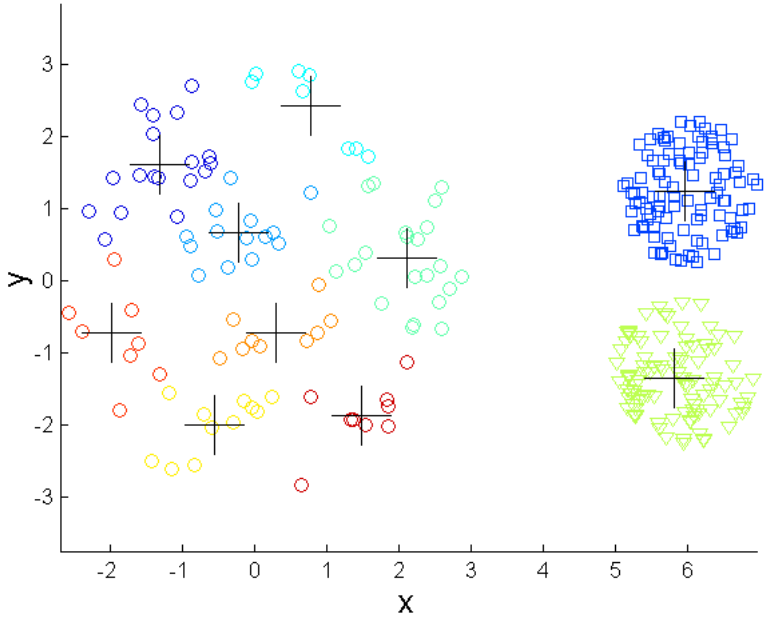
One solution is to use many clusters.

Find parts of clusters, but need to put together.

Overcoming K-means Limitations

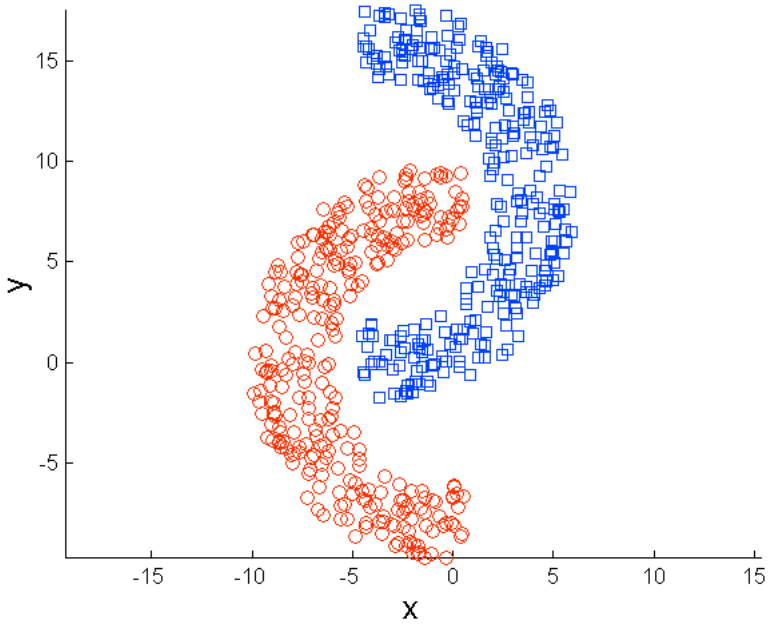


Original Points

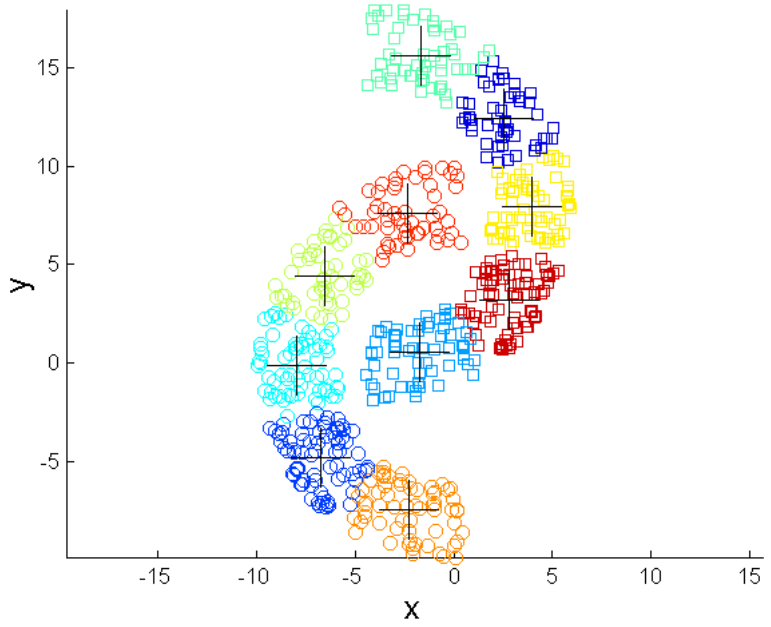


K-means Clusters

Overcoming K-means Limitations



Original Points



K-means Clusters

K-means clustering summary

Advantages

- Items automatically assigned to clusters
- Simple, understandable
- *Relatively efficient*: $O(tkn)$, where n is # objects, k is # clusters, and t is # iterations. Normally, $k, t << n$

Disadvantages

- Must pick number of clusters before hand
- Result can vary significantly depending on initial choice of seeds (number and position)
- All items forced into a cluster
- Too sensitive to outliers
- Not suitable to discover <> clusters (size, density) and with *non-globular shapes*

Cluster Validation

- Cluster Interpretability
- Cluster separation
- Cluster accuracy
- Etc.

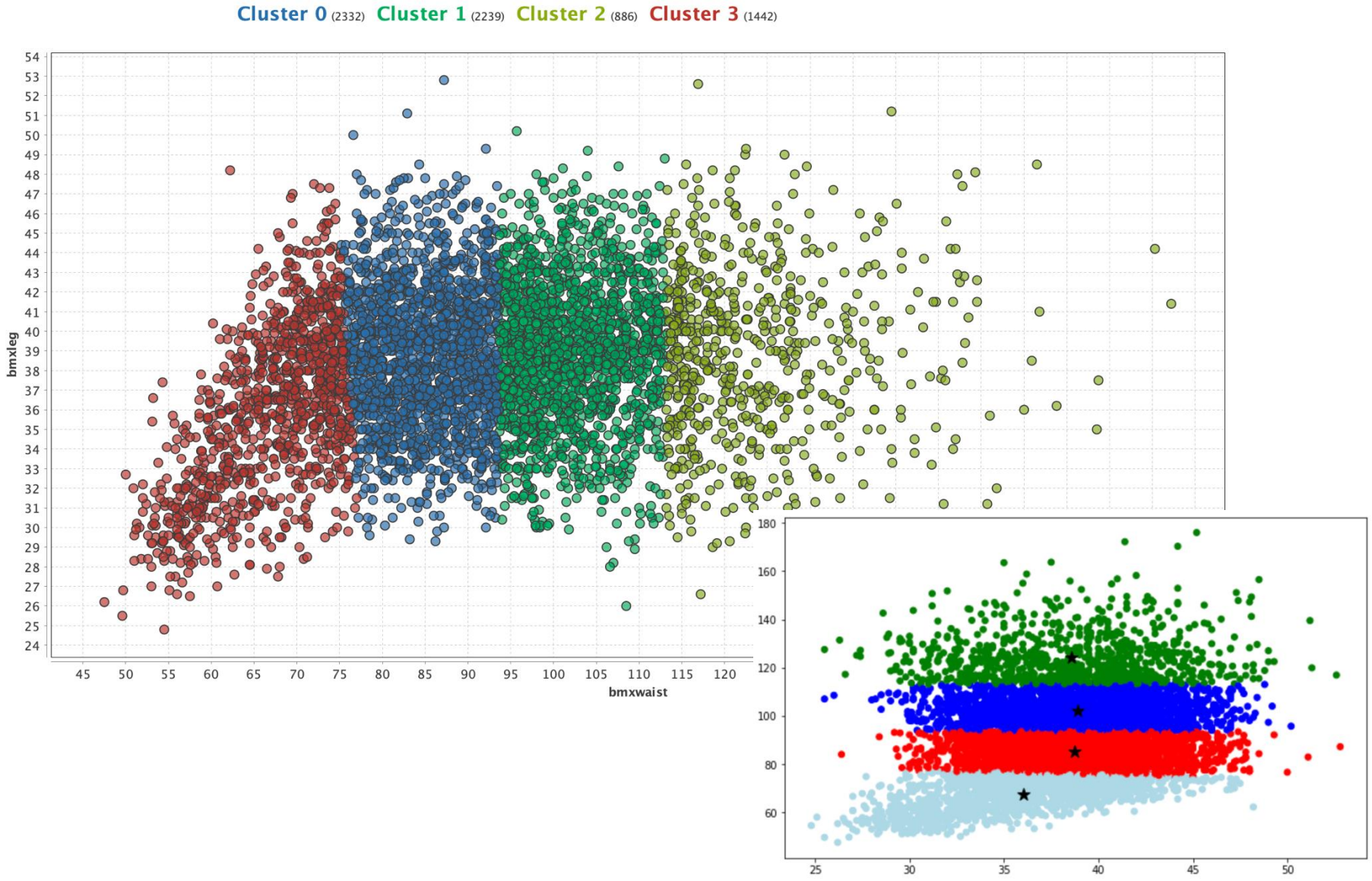
Cluster Interpretability

- Finding clusters is not often an end in itself.
- Once clusters have been detected, other methods must be applied in order to figure out Meaning of clusters.

Cluster Interpretability

- **Goal:** Obtain meaningful and useful clusters
- **Possible solutions**
  - Summary statistics of each cluster
  - Use visualization
  - Distribution compared to the whole population
    - for example,
    - Mean (of each attribute in a cluster) compared to Mean of the population
  - Use dimension reduction
  - Etc.

Cluster Interpretability: 2-Dimension



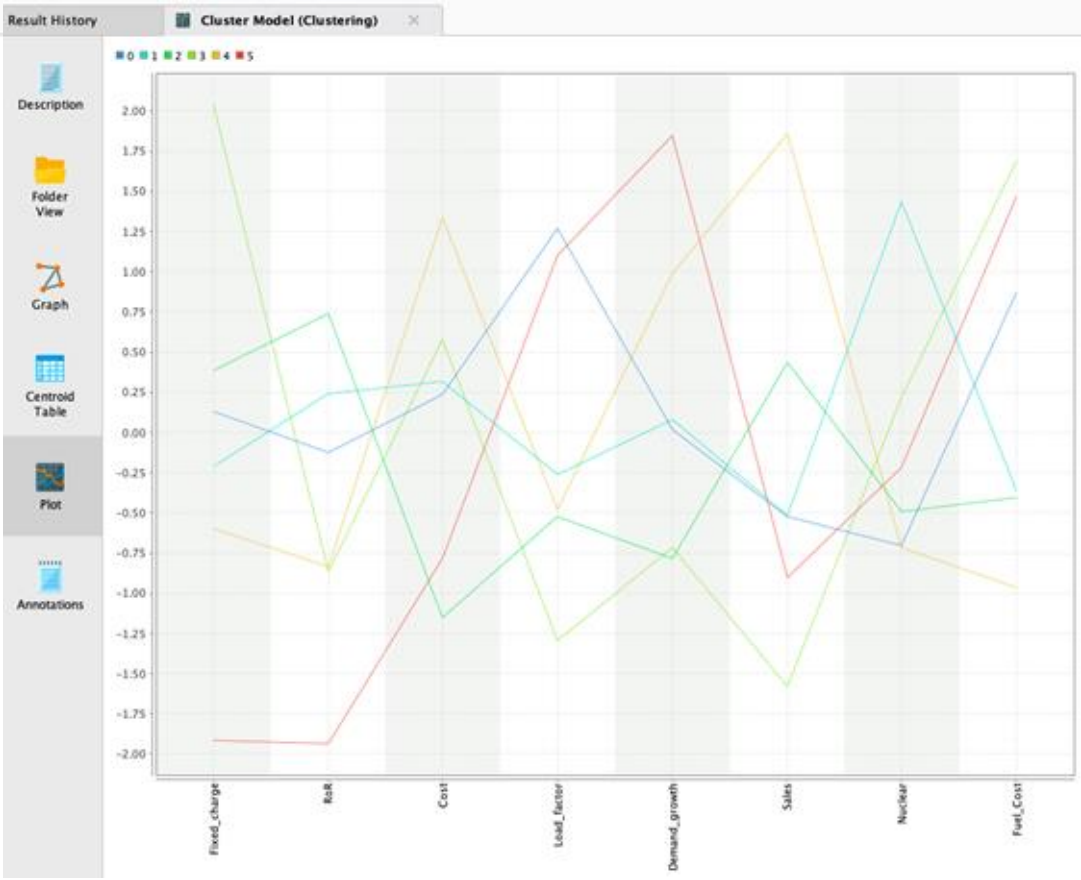
Interpreting K-means results

Result History | Cluster Model (Clustering)

Description
Folder View
Graph
Centroid Table

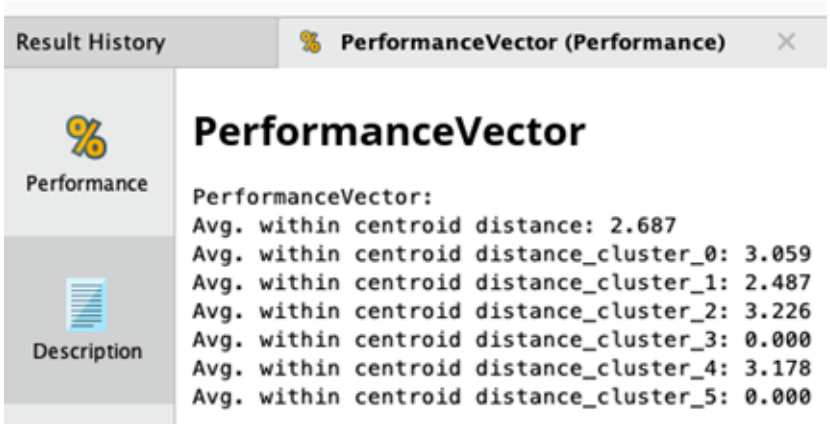
Attribute	cluster_0	cluster_1	cluster_2	cluster_3	cluster_4	cluster_5
Fixed_charge	0.130	-0.212	0.384	2.037	-0.600	-1.919
RoR	-0.123	0.244	0.741	-0.863	-0.833	-1.932
Cost	0.234	0.319	-1.149	0.578	1.339	-0.781
Load_factor	1.269	-0.264	-0.522	-1.295	-0.481	1.103
Demand_gro...	0.019	0.083	-0.783	-0.719	0.992	1.847
Sales	-0.521	-0.515	0.434	-1.581	1.857	-0.901
Nuclear	-0.704	1.436	-0.491	0.214	-0.715	-0.220
Fuel_Cost	0.868	-0.361	-0.407	1.693	-0.966	1.470

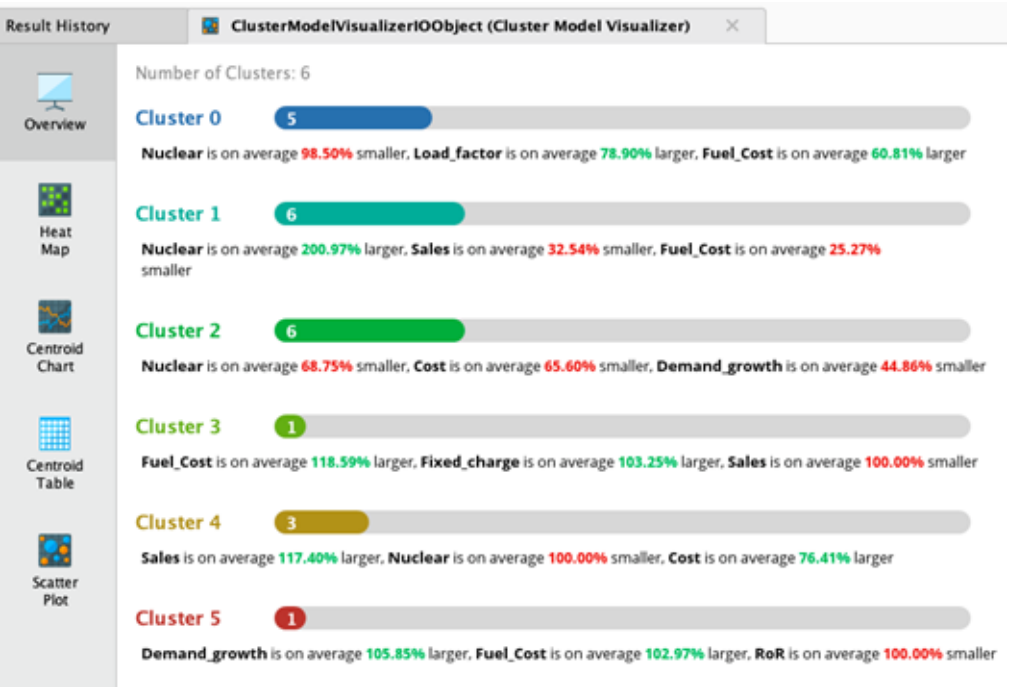
- Centroid Table
- Plot View – Centroids
- Plot View – Observations
- Scatter plot of clusters

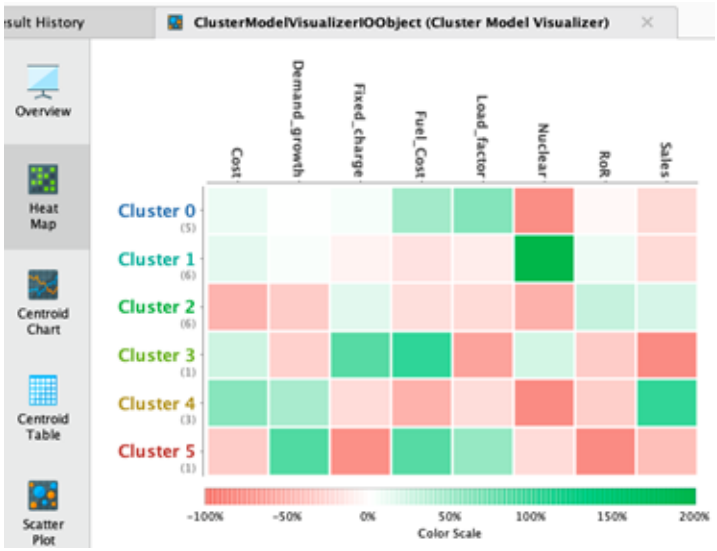


Interpreting K-means performance results

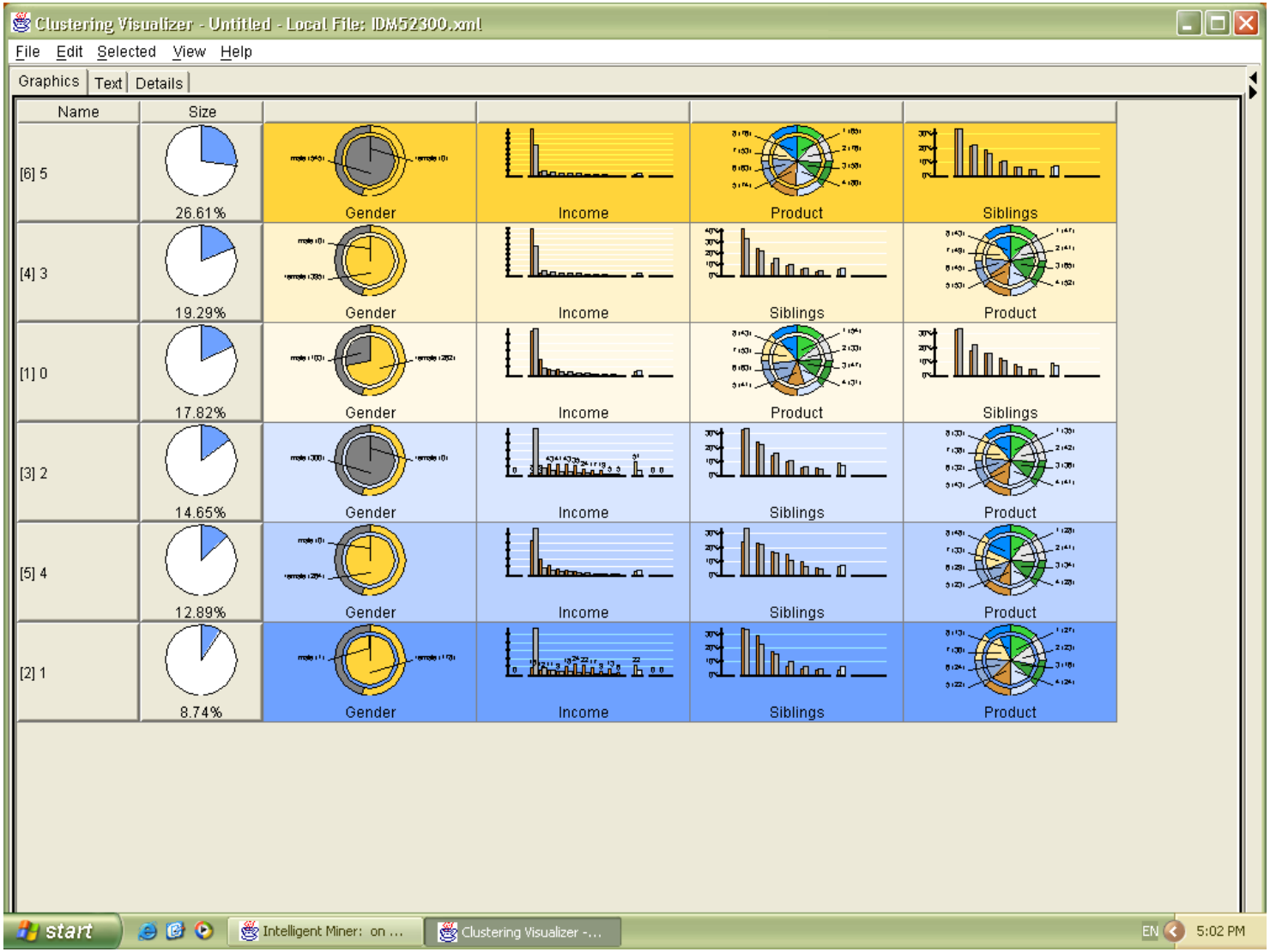
- Performance vector description
- Cluster model visualizer
- Heat map



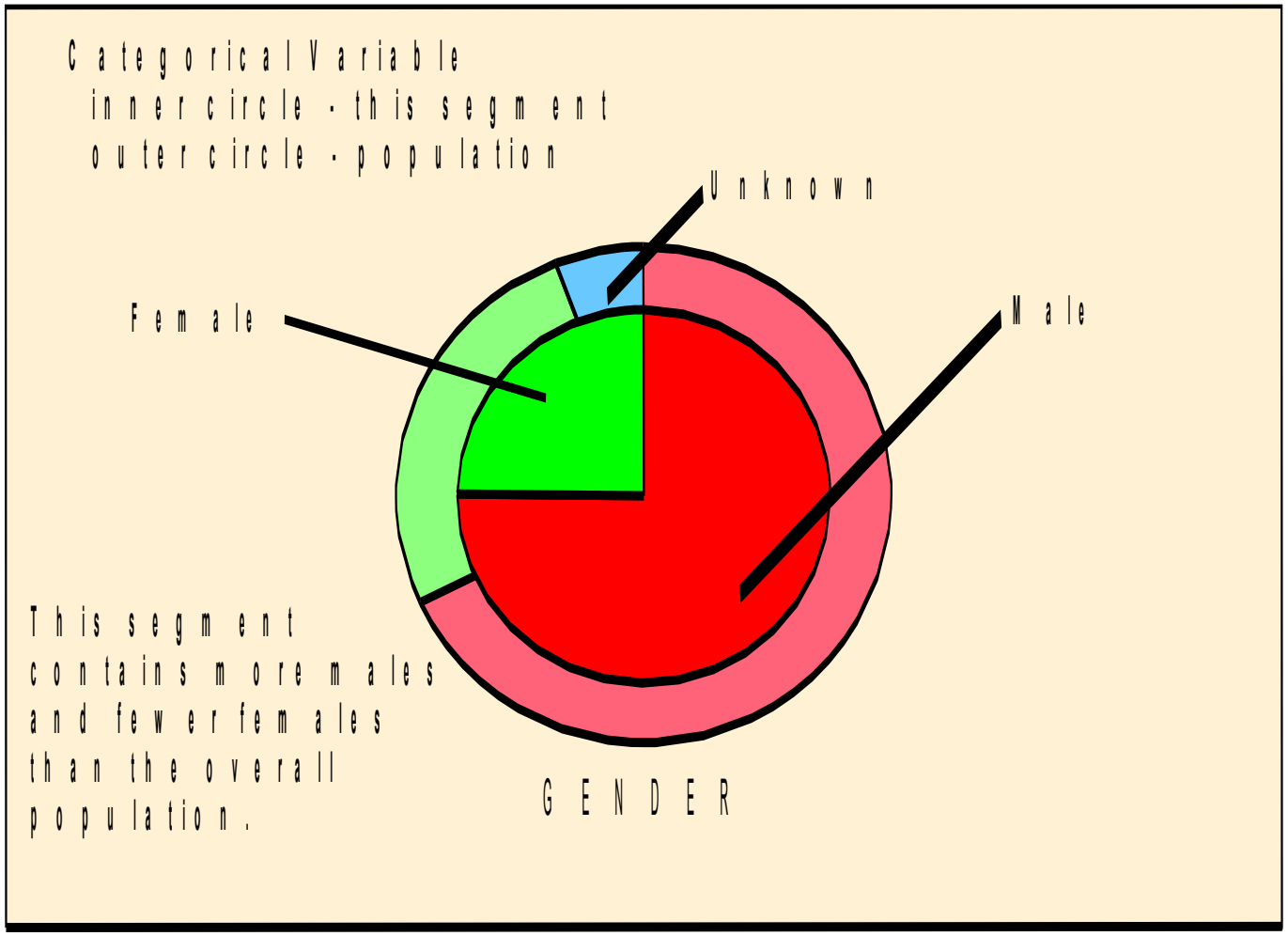




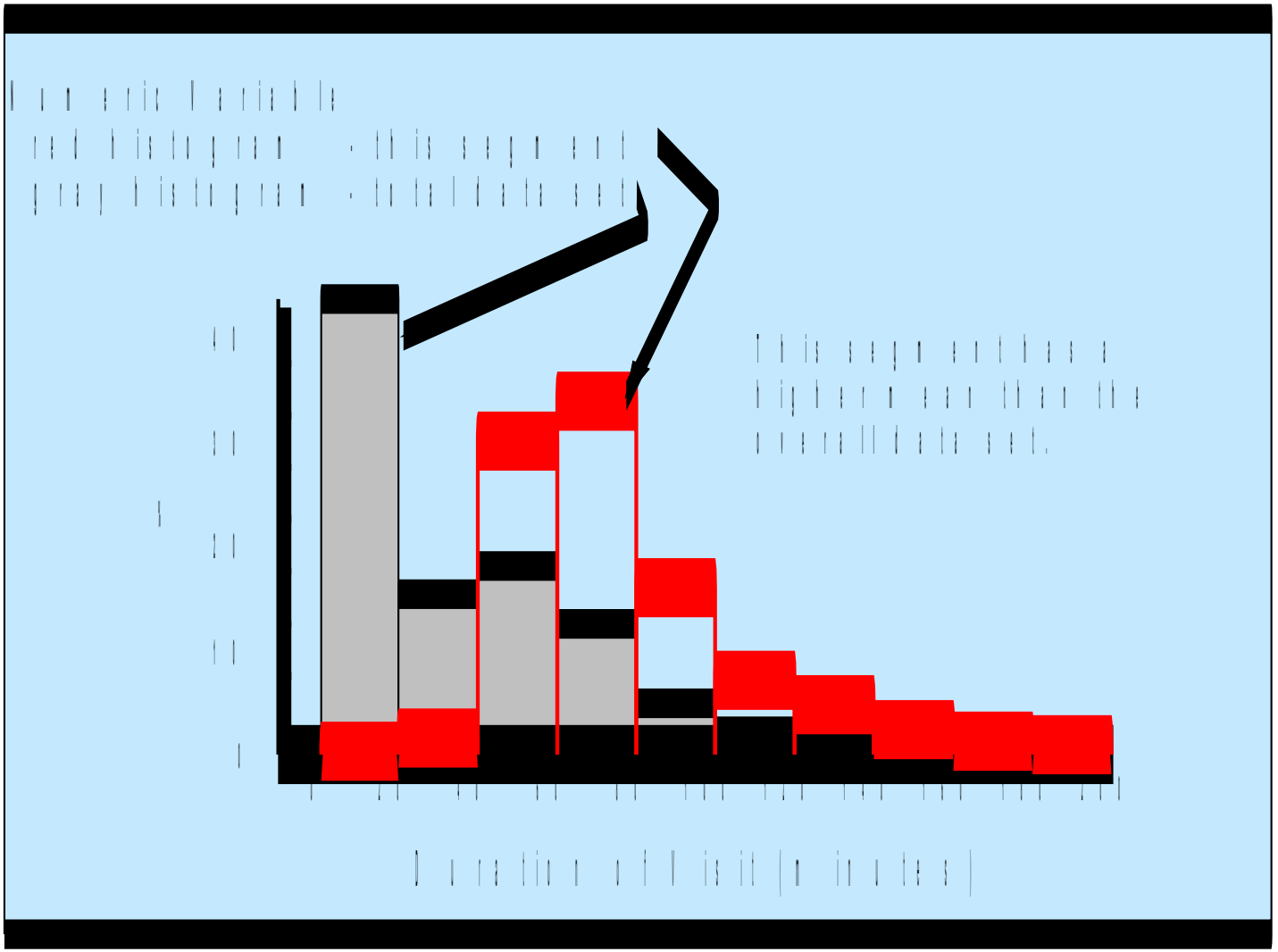
Cluster Interpretability: N-Dimensions



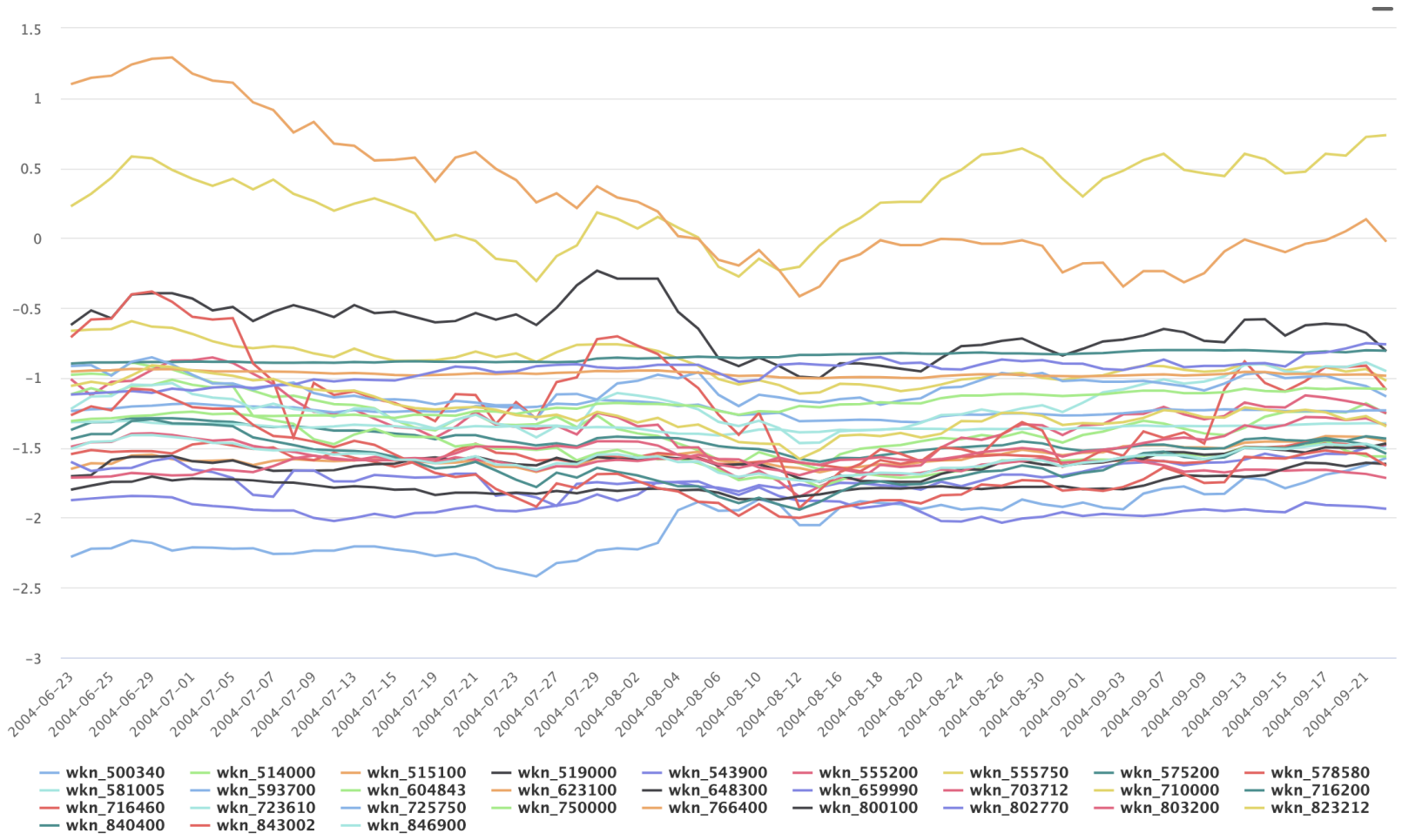
Cluster Interpretability : Categorical variables



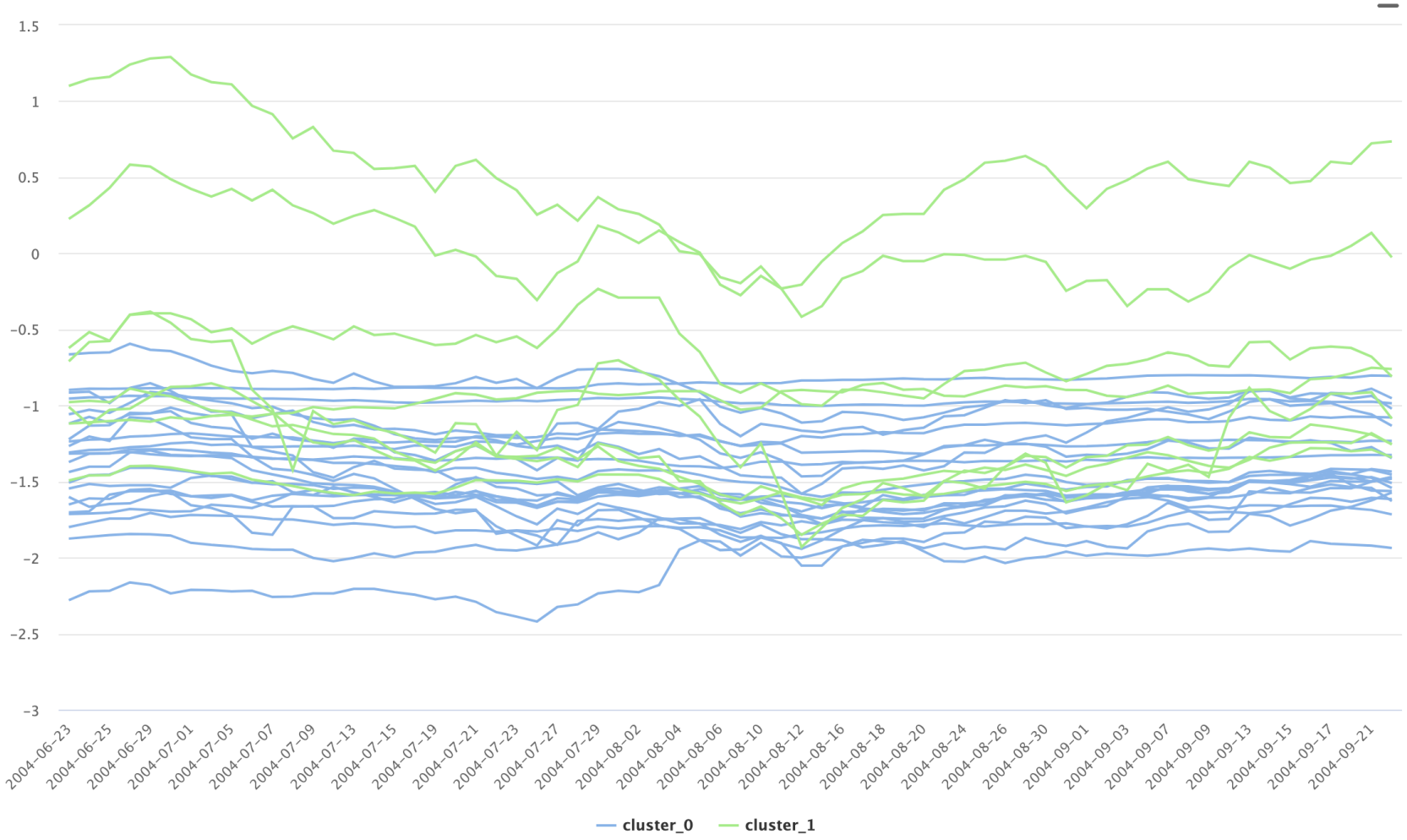
Cluster Interpretability : Numerical variables



Time series data



Cluster Visualization: Time series data



Cluster Separation

Separation – Check ratio of between-cluster variation to within-cluster variation

Company	Fixed_charge	RoR	Cost	Load	Δ Demand	Sales	Nuclear	Fuel_Cost
Arizona	1.06	9.2	151	54.4	1.6	9077	0	0.628
Boston	0.89	10.3	202	57.9	2.2	5088	25.3	1.555
Central	1.43	15.4	113	53	3.4	9212	0	1.058
Commonwealth	1.02	11.2	168	56	0.3	6423	34.3	0.7
Con Ed NY	1.49	8.8	192	51.2	1	3300	15.6	2.044
Florida	1.32	13.5	111	60	-2.2	11127	22.5	1.241
Hawaiian	1.22	12.2	175	67.6	2.2	7642	0	1.652
Idaho	1.1	9.2	245	57	3.3	13082	0	0.309
Kentucky	1.34	13	168	60.4	7.2	8406	0	0.862
Madison	1.12	12.4	197	53	2.7	6455	39.2	0.623
Nevada	0.75	7.5	173	51.5	6.5	17441	0	0.768
New England	1.13	10.9	178	62	3.7	6154	0	1.897
Northern	1.15	12.7	199	53.7	6.4	7179	50.2	0.527
Oklahoma	1.09	12	96	49.8	1.4	9673	0	0.588
Pacific	0.96	7.6	164	62.2	-0.1	6468	0.9	1.4
Puget	1.16	9.9	252	56	9.2	15991	0	0.62
San Diego	0.76	6.4	136	61.9	9	5714	8.3	1.92
Southern	1.05	12.6	150	56.7	2.7	10140	0	1.108
Texas	1.16	11.7	104	54	-2.1	13507	0	0.636
Wisconsin	1.2	11.8	148	59.9	3.5	7287	41.1	0.702
United	1.04	8.6	204	61	3.5	6650	0	2.116
Virginia	1.07	9.3	174	54.3	5.9	10093	26.6	1.306

Cluster Centroids

Cluster	Fixed_charge	RoR	Cost	Load_factor
Cluster-1	0.89	10.3	202	57.9
Cluster-2	1.43	15.4	113	53
Cluster-3	1.06	9.2	151	54.4

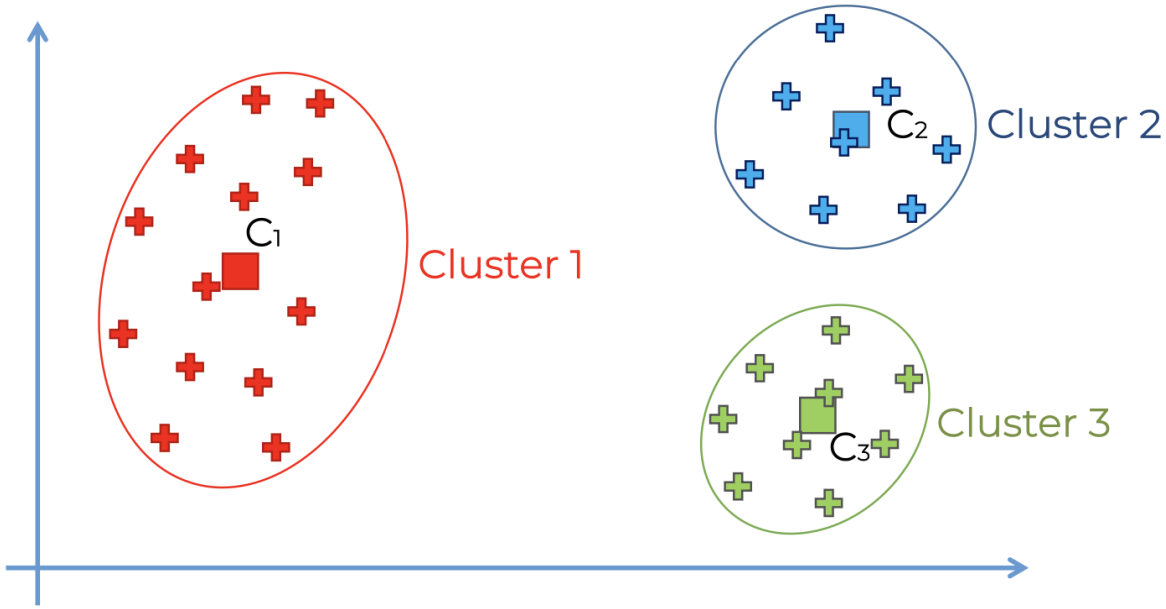
We chose $k = 3$

4 of the 8 variables are shown

Within-Cluster Squared Distance: WCSS

Within Cluster Sum of Squares (WCSS) is the sum of the squared distance between each member of the cluster and its centroid

$$WSS = \sum_{i=1}^{m} (x_i - c_i)^2$$

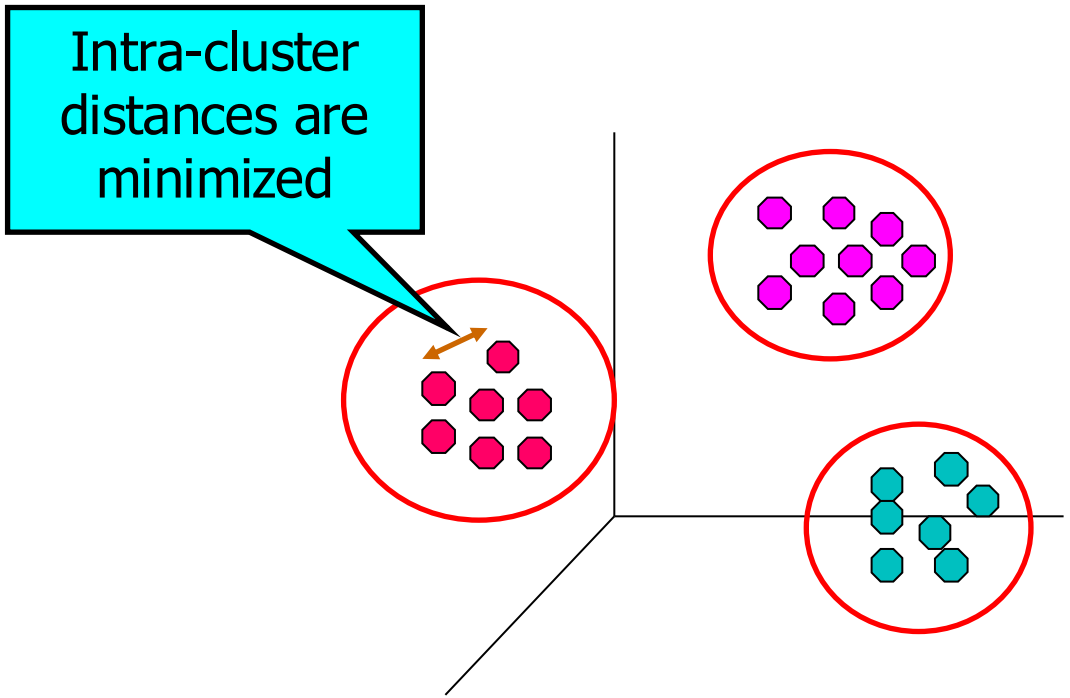


$$\sum_{P_i \text{ in Cluster 1}} \text{distance}(P_i, C_1)^2 + \sum_{P_i \text{ in Cluster 2}} \text{distance}(P_i, C_2)^2 + \sum_{P_i \text{ in Cluster 3}} \text{distance}(P_i, C_3)^2$$

Cluster Distance Performance in RapidMiner

Avg._within_centroid_distance

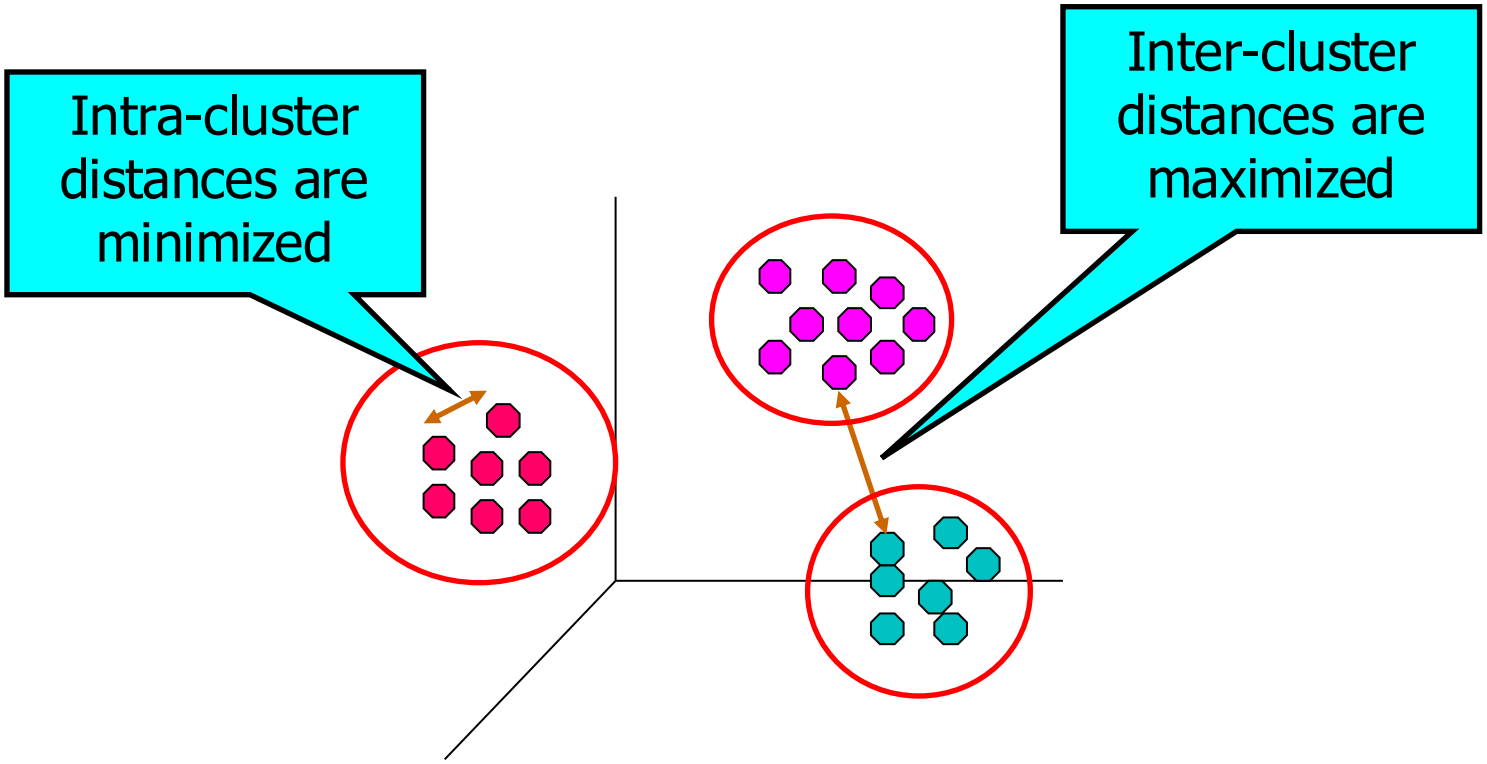
- Average within cluster distance
- Calculated by averaging the distance between the centroid and all examples of a cluster.



Cluster Distance Performance in RapidMiner

Davies–Bouldin index

- The algorithms that produce clusters with low intra-cluster distances (high intra-cluster similarity) and high inter-cluster distances (low inter-cluster similarity) will have a **low Davies–Bouldin** index.
- The algorithm that produces a collection of clusters with the **smallest Davies–Bouldin index** is considered the best algorithm based on this criterion.



Within-Cluster Dispersion

Data summary (In Original coordinates)

Cluster	#Obs	Average distance in cluster
Cluster-1	12	1748.348058
Cluster-2	3	907.6919822
Cluster-3	7	3625.242085
Overall	22	2230.906692

Clusters 1 and 2 are relatively tight, cluster 3 very loose

Conclusion: Clusters 1 & 2 well defined, not so for cluster 3

Next step: try again with k=2 or k=4

Distance Between Clusters

Distance between cluster	Cluster-1	Cluster-2	Cluster-3
Cluster-1	0	5.03216253	3.16901457
Cluster-2	5.03216253	0	3.76581196
Cluster-3	3.16901457	3.76581196	0

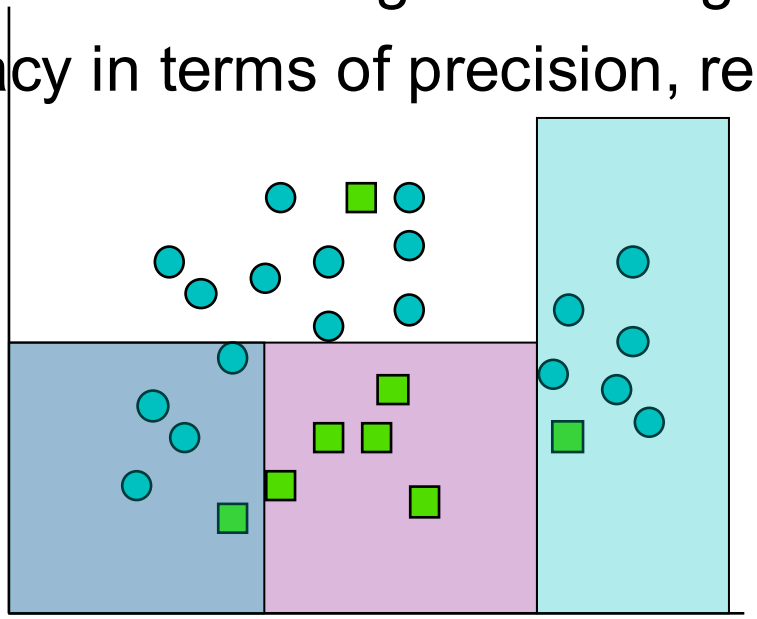
Clusters 1 and 2 are relatively well-separated from each other, while cluster 3 not as much

Use of Labelled Data for Evaluating Cluster Accuracy

Use of ground truth labels

Accuracy– Benchmarking on existing labels

- Accuracy in terms of precision, recall



Summary

- Cluster analysis is an exploratory tool, very useful only when it produces meaningful clusters
- 3 main types of clustering
  - Partitioning clustering is computationally cheap and more stable; requires user to set *k*
  - Hierarchical clustering gives visual representation of different levels of clustering
  - Density-based clustering is able to discover clusters of arbitrary shape
- Other clustering approaches
  - Grid-based approach
  - Model-based
  - Frequent pattern-based
  - Constraint-based
  - Link-based clustering
  - Etc.